

Effective Tax Strategies: It's Not Just Minimization

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September 2014

Abstract: This study examines managerial incentives and practices associated with firms' tax strategy choices, as well as the relative importance of these factors in determining the primary focus of firms' tax strategies. Understanding the determinants of firms' choice of tax strategy is important because the Scholes-Wolfson framework argues that the goal of effective tax planning is to maximize after-tax returns. Therefore, identifying managerial incentives and practices that influence tax strategy choice provides insight into how firms encourage tax planning to improve firm value. I examine two tax strategies: a sustainable tax strategy, which strives to achieve a consistent tax outcome over time, and a minimization tax strategy, which seeks to achieve the lowest possible tax outcome. Controlling for the interdependence of tax strategies, I find that firms are more likely to emphasize a sustainable tax strategy when the CEO's wealth is more sensitive to changes in the firm's stock price and less likely to emphasize sustainability when the firm receives more information from its directors' connections. In contrast, a firm is more likely to concentrate on a minimization tax strategy when the CEO's wealth is more sensitive to changes in stock return volatility, the firm hires a tax expert audit firm for tax services, or the firm receives more information from its directors' connections. Finally, I find that managerial incentives are the most important factors for the choice of tax strategy, followed by the practices of obtaining expert tax advice and information from directors' connections.

Keywords: sustainable tax strategies; minimization tax strategies; managerial incentives and practices

Data Availability: Data used in this study are available from public sources identified in the paper.

Acknowledgements: I am grateful for the comments and insights of my dissertation committee: Tom Omer (Chair), Sean McGuire, Anwer Ahmed, and Dudley Poston. I appreciate helpful comments from Andy Bauer, Anne Beatty, Michael Donohoe, Mike Ettredge, Andrew Finley, Jere Francis, Jeff Hoopes, Tom Kubick, Pete Lisowsky, Lillian Mills, Nate Newton, Ed Outslay, Sonja Rego, Darren Roulstone, and Dan Wangerin and workshop participants at Indiana University, Michigan State University, The Ohio State University, Texas A&M University, the University of Kansas, the University of Illinois at Urbana-Champaign, the University of Missouri, the 2013 Accounting Ph.D. Rookie Recruiting and Research Camp, and the 2014 AAA Annual Meeting, as well as the Iowa Tax Readings Group and the University of Arizona Tax Readings Group. I also thank Bree Josefy for her suggestions and Frances Tice for sharing the director connectedness data.

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I. Introduction

Recent empirical and anecdotal evidence suggests that a firm's tax strategy consists of at least two recognizable dimensions, sustainability and minimization (e.g., Deloitte LLP 2013; KPMG LLP 2007; McGuire et al. 2013). The dimension of a firm's tax strategy that focuses on sustainability strives to achieve a consistent tax outcome over time (i.e., a narrow range of effective tax rates (ETRs)), with less regard for that tax outcome's level (McGuire et al. 2013; Neuman et al. 2013). In contrast, the dimension of a firm's tax strategy that focuses on minimization seeks to reduce the firm's tax burden and achieve the lowest possible tax outcome (i.e., a low ETR). Despite the presence of multiple dimensions, prior research generally focuses on firms' tax avoidance with the implicit assumption that *all* firms emphasize minimization (Shackelford and Shevlin 2001; Hanlon and Heitzman 2010); however, only a subset of firms are able to achieve low ETRs (Dyreng et al. 2008), implying that there are alternative approaches to maximizing after-tax returns. I extend this line of research by investigating why firms choose to concentrate more on sustainability or minimization. Specifically, I examine the managerial incentives and practices associated with firms' tax strategy choices, as well as the relative importance of these factors in determining the primary focus of firms' tax strategies.¹

Tax strategy can be defined as a pattern of decision-making and actions with respect to tax planning that maximizes after-tax returns and firm value (Scholes et al. 2009). Understanding the determinants of firms' tax strategy choices is important because the Scholes-Wolfson framework argues that the goal of effective tax planning, the foundation of tax strategy, is to maximize after-tax returns (Scholes et al. 2009). By examining the variation in tax outcomes and the determinants and consequences of tax avoidance, prior research has sought to explain observed tax outcomes (e.g., ETRs, book-tax differences, and the incidence of tax sheltering). In contrast, this study examines firms' tax decisions from a different perspective, their choice of tax *strategy*, which precedes observed tax outcomes.

¹ Following prior research, I acknowledge that sustainability and minimization are dimensions of the overarching tax strategy. However, for ease of exposition, I refer to them as sustainable or minimization tax strategies.

Prior research identifies two different dimensions of firms' tax strategies, improving the consistency of the tax outcome (sustainability) and reducing cash taxes paid (minimization). Non-tax considerations are important factors for tax planning (e.g., Scholes et al. 2009), and the literature indicates that firms may prefer sustainability to avoid tax-related surprises (Armstrong et al. 2012; Schmidt 2006; TEI 2005) and balance tax avoidance and financial reporting costs (Mills 1998; Mills and Newberry 2001). Firms benefit from emphasizing sustainability in three primary ways: improved firm value through lower earnings and cash flow volatility (e.g., Barth et al. 1999; Myers et al. 2007; Rountree et al. 2008), reduced likelihood of unfavorable financial reporting effects (Mills 1998; Scholes et al. 2009), and decreased exposure to the reputation costs of negative tax-related press (Graham et al. 2014; Hanlon and Slemrod 2009). However, by investing in tax planning that produces more predictable tax outcomes, these firms could be paying more tax than legally required (Deloitte LLP 2013; TEI 2005).

On the other hand, prior research suggests that managers are often incentivized to minimize tax outcomes (Phillips 2003; Robinson et al. 2010; Armstrong et al. 2012), and anecdotal and empirical evidence indicates that some firms report extremely low ETRs (e.g., Dyreng et al. 2008; Schumpeter 2012; Kocieniewski 2011; Duhigg and Kocieniewski 2012). Firms focusing on minimization benefit from the increased cash flow of lower explicit tax burdens (Hanlon and Heitzman 2010; Maydew 1997; Mills 1998; Phillips 2003); however, low ETRs also increase the risk of audit by revenue authorities, possibly resulting in penalties and fines for disallowed tax avoidance (Ayers et al. 2009; Cloyd 1995; Cloyd et al. 1996; Mills 1998; Mills and Sansing 2000; Schumpeter 2012). Furthermore, managers are concerned about the reputation costs associated with tax minimization (Graham et al. 2014).

Prior literature indicates that all commonly-observed strategies are capable of allowing firms to earn acceptable returns (Porter 2004); thus, although both tax strategies have costs and benefits, it is important to note that *either* tax strategy can maximize after-tax firm value. However, firms' unique situations ultimately determine their strategic choice (Porter 2004; Miles et al. 1978; Hambrick 1983). Therefore, I investigate the determinants of firms' tax strategy choices to improve our understanding of firms' decisions to emphasize sustainable or minimization tax strategies.

Managerial incentives and practices shape managers' values and facilitate decision-making that is consistent with shareholders' interests (Jensen and Meckling 1976). Mills et al. (1998) assert that investments in tax planning will vary with managements' practices and incentives; therefore, I expect the choice of tax strategy to also depend upon firms' incentives and practices. Accordingly, I examine the association between tax strategy choice and three managerial incentives and practices: providing equity incentives, obtaining expert tax advice, and information transferred through directors' connections.

Hanlon and Heitzman (2010) argue that managerial incentives should align managers' and shareholders' interests with respect to tax planning decisions. Furthermore, prior research provides evidence that shareholders offer managerial incentives to encourage tax planning (Armstrong et al. 2012; Desai and Dharmapala 2006; Phillips 2003; Rego and Wilson 2012). However, equity incentives encourage different types of behavior; thus, I expect these incentives to differentially impact tax strategy choice. Increasing the sensitivity of the manager's wealth to changes in the stock price (*delta*) encourages the manager to increase the stock price without increasing volatility. Firms with more persistent earnings exhibit higher stock prices (e.g., Barth et al. 1999; Kasnik and McNichols 2002; Myers et al. 2007), and firms with more sustainable tax strategies exhibit more persistent earnings (McGuire et al. 2013). Therefore, if managers are incentivized to increase the stock price, they may be more likely to focus on sustainability. On the other hand, increasing the sensitivity of the managers' wealth to stock return volatility (*vega*) encourages the manager to make relatively more risky investments. A minimization tax strategy will likely require the manager to undertake aggressive tax avoidance activities or uncertain tax positions (Hanlon et al. 2007; Rego and Wilson 2012), which suggests that, if managers are rewarded for risk (stock return volatility), they may be more likely to emphasize minimization.

In addition to managerial incentives, I also consider two managerial practices that facilitate the spread of tax planning ideas, obtaining expert tax advice and transferring information through directors' connections. Expert tax advice is a valuable source of tax planning information because accounting firms invest in training and knowledge to serve their tax clients' needs (Ferguson et al. 2003; Solomon et al. 1999). Tax experts offer information about recent tax planning innovations, as well as industry-specific

tax avoidance activities, which can improve their clients' abilities to emphasize minimization (Deloitte Development LLC 2013; Ernst & Young 2013; PricewaterhouseCoopers 2013); however, practitioners also market tax planning activities with sustainable tax outcomes (Deloitte LLP 2013; KPMG LLP 2007; PwC 2008), suggesting expert tax advice could also be associated with sustainable tax strategies.

Relationships among directors form an extensive network across which information, including corporate policies, is shared (e.g., Gulati and Westphal 1999; Bizjak et al. 2009), and these connections are sources of tax planning information. Prior research has shown that tax shelter use spreads through board interlocks (Brown 2011) and firms with interlocking directors pay similar amounts of tax (Brown and Drake 2014). Furthermore, anecdotal evidence indicates that tax directors meet regularly with boards to discuss tax issues (ATA 2014); thus, directors communicate directly with the individuals responsible for implementing tax strategy. Directors' specific connections will determine the type of tax planning information they have access to (i.e., predictable (innovative) tax planning opportunities which facilitate sustainability (minimization)). Thus, depending on the information they receive, information transferred through directors' connections could be associated with either sustainable or minimization tax strategies.

To investigate whether managerial incentives and practices are determinants of firms' tax strategy choices, I examine the association between equity incentives, expert tax advice, and information transferred through directorships and the likelihood of adopting sustainable or minimization tax strategies. I classify a firm as emphasizing a sustainable tax strategy if its coefficient of variation of cash ETRs, measured from t to $t-4$ (McGuire et al. 2013), is in the lowest quintile for the year and as emphasizing a minimization tax strategy if its three-year cash ETR, measured from t to $t-2$ (Dyreng et al. 2008), is less than or equal to 20 percent.² I measure equity incentives using *delta* and *vega* (Core and Guay 1999; Guay 1999), expert tax advice using audit firm tax expertise (McGuire et al. 2012), and information transferred through directorships using director connectedness (Newman 2010; Omer et al. 2012, 2014).³

² My inferences remain unchanged if I use the five-year cash ETR to define a minimization tax strategy.

³ To measure information transferred by directors comprehensively, I use social network analysis and a vector of four director centrality measures following the methodology of Newman (2010) rather than board interlocks.

Using a sample of firm-year observations at the intersection of the Compustat, CRSP, Execucomp, Audit Analytics, and Risk Metrics databases from 2000-2010, I investigate firms' tendencies to focus on one of two tax strategies using seemingly unrelated bivariate probit regression, a methodology that accounts for the simultaneous determination of the decision to focus on one dimension of tax strategy and forego emphasizing the other. Controlling for firm characteristics that could affect tax planning resources and/or opportunities, my results indicate that, when the CEO's wealth is more sensitive to changes in stock price (i.e., larger *delta*), the firm is more likely to concentrate on sustainability; however, firms that incentivize their CEOs to make riskier decisions (i.e., larger *vega*) are more likely to emphasize minimization. In addition, I find that firms that hire a tax expert audit firm for tax services are more likely to focus on minimization, suggesting that these firms utilize tax experts' knowledge of new tax planning opportunities. Finally, I find that firms with well-connected directors are less likely to focus on sustainability and more likely to adopt a minimization tax strategy, implying that directors share more information about tax minimization opportunities than activities that produce consistent tax outcomes.

To examine the relative importance of managerial incentives and practices in determining firms' tax strategies, I include all variables of interest in the analysis simultaneously and find that, for both tax strategies, equity incentives are the most important factor for explaining tax strategy choice, followed by expert tax advice, and finally, information from directors' connections. Thus, equity incentives are not only associated with observed tax outcomes, as documented by prior literature, but more importantly, represent one of the primary determinants of tax strategy choice. In addition, to further control for the effect of tax planning opportunities on tax strategy choice, I replicate my results using a sample of sustainable and minimization tax strategy firms matched on size within industry and year and find that managerial incentives and practices remain significant determinants of firms' tax strategy choices.

This study makes three primary contributions to the literature. First, I extend the literature that examines the determinants of corporate tax avoidance by investigating firms' choice of tax strategy, by providing evidence that managerial incentives and practices are associated with firms' tax strategy choices, and by examining the relative importance of these factors for the choice of tax strategy.

Although prior literature has investigated some of these characteristics and their association with tax avoidance and tax sheltering activities, it has not analyzed these incentives and practices in the context of how they influence tax strategy choice, a decision that is a precursor to observed tax outcomes.

Second, despite prior literature's identification of numerous factors and firm characteristics that are associated with firms' tax avoidance, little of the variation in tax outcomes has been explained (Hanlon and Heitzman 2010; Shevlin 2007) and additional research is needed to explain why all firms do not minimize taxes (Maydew 2001). By classifying firms based on their tax strategies and identifying managerial incentives and practices that affect tax strategy choice, this study provides additional explanations for the observed variation in tax outcomes and answers the calls of prior research to specifically examine those firms that do not minimize taxes. Finally, by identifying managerial incentives and practices associated with the choice of tax strategy, this study provides insight into how firms encourage tax planning that improves firm value.

II. Background on Tax Strategy and Its Dimensions

The management literature defines a strategy as an observable pattern in decision-making (Hambrick 1981, 1983; Miles and Snow 1978; Mintzberg 1978). The goal of any strategy can be achieved using a variety of approaches; thus, all firm strategies are inherently multi-faceted with each approach capable of maximizing value; however, managers implement a strategy that complements the firm's competitive advantage (Hambrick 1983; March 1991; Miller 1987; Porter 2004; Zahra and Covin 1993). Consistent with this notion, the Scholes-Wolfson framework argues that, because of non-tax costs, effective tax planning is not always achieved by tax minimization (Scholes et al. 2009), implying that different, yet equally effective, approaches to tax strategy exist. Prior research supports this argument. Dyreng et al. (2008) indicate that tax avoidance does not always improve market value because reducing explicit taxes may increase implicit taxes, or non-tax costs. Similarly, Robinson et al. (2010) recognize that lower ETRs do not always imply effective tax planning, and Phillips (2003) states that a low ETR represents tax savings, but does not necessarily indicate that after-tax returns have been maximized. Furthermore, although some firms achieve substantial tax avoidance (e.g., Dyreng et al. 2008), many

firms reduce their tax burdens by relatively small amounts. Thus, the literature provides a strong basis for concluding that there are multiple tax strategies that can maximize after-tax returns.

The strategy literature classifies firms into typologies by identifying the most salient strategies. Following this approach, I focus on sustainability and minimization because practitioners commonly discuss these strategies and firms following either strategy are easily identified empirically. Furthermore, firms explicitly state their emphasis on sustainable or minimization tax strategies.⁴ A sustainable tax strategy consists of recurring (long-term) tax planning techniques that produce consistent tax outcomes over time (e.g., cost recovery of expenditures and unrepatriated foreign earnings from stable countries with undisputed tax laws). Firms implementing this tax strategy are mindful of the variability of tax outcomes, but less concerned about the tax outcome's level (McGuire et al. 2013; Neuman et al. 2013). In contrast, a minimization tax strategy strives to achieve the lowest possible tax outcome using a combination of predictable and less certain, recurring and one-time tax planning opportunities (Mills et al. 1998). Firms implementing this tax strategy are concerned about the level of taxes being paid and pursue a much wider range of tax avoidance activities (e.g., cost recovery of expenditures, unrepatriated earnings from a variety of foreign operations, research and experimentation tax credits, and transfer pricing).

Empirical and anecdotal evidence confirms the existence of both tax strategies. With respect to sustainability, Mills (1998) and Mills and Newberry (2001) provide evidence that firms balance tax avoidance and financial reporting costs; prior research and survey evidence indicate that firms prefer to avoid tax-related surprises (Schmidt 2006; Armstrong et al. 2012; TEI 2005); and McGuire et al. (2013) find that some firms have little variability in their tax outcomes. In addition, practitioners advertise tax planning activities with sustainable outcomes because these tax strategies increase long-term value

⁴ For example, Unilever employs tax principles which “illustrate good corporate practice in the area of tax management and tax transparency, balancing the interests of our various stakeholders, [including] the governments and communities in the countries in which we operate (Unilever 2013).” Rio Tinto states that it “pursues a tax strategy that is principled, transparent, and sustainable in the long term (Rio Tinto 2013).” In contrast, Anne Eisele stated that GE is “committed to complying with tax rules and paying all legally obliged taxes; [however,] we have a responsibility to our shareholders to legally minimize our costs (Kocieniewski 2011).” Google stated that it operated “in a manner consistent with all applicable laws” and was “satisfying [its] fiduciary duties to [its] stockholders” and “[its] legal obligations in each of the countries in which [it] operate[s] (Google, Inc. 2014, p.59).”

(Strategic Capital Partners, LLC 2012; Deloitte LLP 2013; KPMG LLP 2007; PwC 2008). However, prior research also provides evidence of minimization tax strategies. Managers are often incentivized to reduce tax outcomes (Phillips 2003; Robinson et al. 2010; Armstrong et al. 2012) because it increases income and, in some instances, encourages behavior that mitigates incentive problems (Scholes et al. 2009). Furthermore, there is substantial evidence that some firms are able to achieve low ETRs (e.g., Dyreng et al. 2008; Schumpeter 2012; Kocieniewski 2011; Duhigg and Kocieniewski 2012).

There are costs and benefits associated with both sustainability and minimization. Firms focusing on sustainability experience low year-to-year variability in tax outcomes (Deloitte LLP 2013; KPMG LLP 2007; PwC 2008) and are less likely to have tax-related earnings surprises (Armstrong et al. 2012; Schmidt 2006; TEI 2005) or unfavorable financial reporting effects (Mills 1998; Scholes et al. 2009). These firms also exhibit more persistent earnings (McGuire et al. 2013), improving firm value through consistent earnings' strings (e.g., Barth et al. 1999; DeAngelo et al. 1996; Kasznik and McNichols 2002; Mills and Newberry 2001; Myers et al. 2007). These characteristics have been linked to firm value (see Dechow et al. (2010) for a review of the literature), but practitioners' also promote adopting a sustainable tax strategy to reduce exposure to negative tax-related press and the associated reputation costs (Deloitte LLP 2013; KPMG LLP 2007; PwC 2008), a significant concern for firms (Graham et al. 2014; Hanlon and Slemrod 2009). However, sustainability is not without costs. By investing in tax planning that produces more predictable outcomes, these firms may be paying more tax than legally owed (Deloitte LLP 2013; McGuire et al. 2013; TEI 2005), reducing cash available for investment.⁵ Furthermore, after adopting this strategy, these firms may be pressured to maintain consistent outcomes because sudden volatility can be viewed as a signal of poor management (KPMG LLP 2007).

Firms emphasizing minimization benefit from the increased income and cash flow of lower explicit tax burdens (Hanlon and Heitzman 2010; Maydew 1997; Mills 1998; Phillips 2003), which can be invested in value-increasing projects (Armstrong et al. 2012; Ayers et al. 2009; Schumpeter 2012).

⁵ On average, firms following sustainable tax strategies in my sample pay higher taxes than firms following minimization tax strategies (average three-year cash ETR of 28.7 percent versus 11 percent).

These firms are not as sensitive to the certainty or timing of tax activities' expected outcomes; thus, they draw upon a wider range of tax planning options. However, low ETRs increase the risk of audit, possibly resulting in tax benefit-negating fines and penalties (Ayers et al. 2009; Cloyd 1995; Cloyd et al. 1996; Mills 1998; Mills and Sansing 2000; Schumpeter 2012). Scholes et al. (2009) also warn that, if non-tax costs are not considered, minimization can be an ineffective tax strategy (Deloitte LLP 2013; Mills and Newberry 2001; Phillips 2003). Tax minimization can also create significant reputation costs. The idea of companies paying their fair share became more widespread during the latest recessionary period (Deloitte LLP 2013; Ernst & Young 2012; Hanlon and Slemrod 2009), leading firms to seek to avoid negative attention, and detrimental consequences, related to their tax strategies (Graham et al. 2014).⁶

I expect firms to emphasize one tax strategy because the management literature suggests that it is almost impossible for firms to successfully implement a dual strategy (March 1991). Firms' stakeholders create policies and managers take actions that depend on the strategic dimension being pursued (Miles and Snow 1978). The sets of viable policies and actions for each dimension of a strategy are unlikely to overlap substantially, which usually prohibits emphasizing multiple dimensions simultaneously. Thus, based on prior literature, and the different costs and benefits of, and tax planning actions afforded by, each tax strategy, I expect firms to emphasize *either* a sustainable or minimization tax strategy and to seldom have the capability to implement both dimensions of tax strategy.⁷

III. Hypothesis Development

Managerial incentives and practices are used to ensure that managers make decisions consistent with stakeholders' interests (Jensen and Meckling 1976). Because they are intended to affect decision-making, these factors likely influence tax strategy choice, a notion that prior research supports. Dyreng et al. (2008) argue that management's actions determine a firm's tax strategy, and Mills et al. (1998) assert that investments in tax planning will vary with management's incentives and practices. Thus, I examine

⁶ For example, Starbucks found the backlash to its legitimate U.K. tax avoidance so harmful to sales that it elected to forgo an allowed deduction and pay \$30.8 million in taxes in 2013 and 2014 (BBC 2013).

⁷ Consistent with prior literature, I find a small sample of firms (187 observations) that emphasize both sustainability and minimization. I conduct additional analyses to provide insights into their ability to focus on both tax strategies.

three managerial incentives and practices – providing equity incentives, obtaining expert tax advice, and information transferred through directorships – and their association with firms’ tax strategy choices.⁸

Equity Incentives

According to principal-agent theory, shareholders should structure compensation contracts and employ incentives to link managers’ wealth to firm performance (Diamond and Verrecchia 1982; Holmstrom 1979; Jensen and Meckling 1976). Consistent with this theory, Core and Guay (1999) find that firms provide equity incentives to target agency conflicts and influence decision-making. Incentives are designed to improve firm performance, and because tax strategy affects profitability (Scholes et al. 2009), the compensation contract should inform managers about the level of effort to devote to tax planning (Hanlon and Heitzman 2010; Slemrod 2004) and align the interests of managers and shareholders with respect to tax strategy decisions (Armstrong et al. 2012; Phillips 2003; Rego and Wilson 2012).⁹ Prior research supports this assertion and concludes that equity incentives and compensation (Armstrong et al. 2012; Desai and Dharmapala 2006; Phillips 2003; Rego and Wilson 2012), as well as specific details of managers’ compensation contracts (Robinson et al. 2010; Powers et al. 2013), are associated with firms’ tax avoidance.

Firms manage and rely primarily on two types of equity incentives – the sensitivity of the manager’s wealth to changes in the stock price (*delta*) and the sensitivity of the manager’s wealth to stock return volatility (*vega*). Increasing *delta* encourages managers to improve firm value by raising the stock price (Guay 1999). A number of earnings quality characteristics have been linked to higher stock prices, but one aspect, earnings persistence, is associated with sustainability. Firms with more persistent or predictable earnings exhibit higher stock prices (e.g., Barth et al. 1999; DeAngelo et al. 1996; Kasznik and McNichols 2002; Myers et al. 2007), and firms with more sustainable tax strategies exhibit more

⁸ Firm characteristics will influence available tax planning opportunities, but I focus on incentives and practices because my objective is to explore factors beyond firms’ innate characteristics that affect tax strategy choice. I include a vector of control variables to account for the impact of firm characteristics on tax strategy choice.

⁹ The compensation contract will not *explicitly* state which tax strategy to choose or how the executive’s time should be allocated; however, it will outline evaluation metrics. Thus, the compensation contract will influence the manager’s tax strategy choice as he seeks to make decisions that maximize his wealth.

persistent earnings (McGuire et al. 2013), suggesting that this tax strategy is related to higher firm performance. Furthermore, risk-averse managers will invest in projects with more predictable outcomes (Smith and Stulz 1985); therefore, when rewarded for increasing the stock price, managers may be more likely to emphasize sustainability. Thus, I expect the sensitivity of the manager's wealth to changes in the stock price (*delta*) to increase the likelihood that a firm engages in a sustainable tax strategy.

Compensating managers for improving firm performance reduces some agency conflicts, but risk-averse managers will avoid valuable, but uncertain, projects (Smith and Stulz 1985). However, by rewarding managers for stock return volatility (*vega*), shareholders can encourage broader investment preferences (e.g., Chava and Purnanandam 2010; Coles et al. 2006; Guay 1999). A firm may be more likely to emphasize minimization when incentive structures encourage risk-taking because, to achieve the lowest possible tax outcome, managers are more likely to undertake risky tax avoidance activities (Desai and Dharmapala 2006; Hanlon et al. 2007; Rego and Wilson 2012) with more volatile tax outcomes, and ultimately, stock returns. In addition, Rego and Wilson (2012) find that the sensitivity of the manager's wealth to stock return volatility is associated with tax avoidance, suggesting a positive relation between incentives to engage in riskier activities and minimization. Furthermore, Guenther et al. (2013) find that firms with more volatile cash ETRs have higher future stock return volatility, suggesting that managers of firms with sustainable tax strategies are unlikely to be rewarded for volatile stock returns. Thus, I expect the sensitivity of the manager's wealth to stock return volatility (*vega*) to increase the likelihood that a firm engages in a minimization tax strategy.

I state my first hypothesis in the alternative form:

H1: *A firm will be more likely to implement a sustainable tax strategy (minimization tax strategy) as its manager's wealth becomes more sensitive to changes in stock price (stock return volatility).*

Managerial Practices

Managerial incentives are important decision-making factors because they align shareholders' and managers' interests and encourage certain types of decisions. Although likely second-order effects relative to incentives, managerial practices may also influence tax strategy choice because they are put

into place by the firm and represent key sources of tax planning information that aid in implementing a tax strategy. I examine two practices – expert tax advice and information transferred among directors.

Expert Tax Advice

Theory suggests that decision-makers solicit advice to improve the likelihood of making an optimal decision (Bonaccio and Dalal 2006; Harvey and Fischer 1997; Yaniv 2004a, b), and consistent with theory, prior research finds that employing professional advice improves decision accuracy (Gardner and Berry 1995; Snizek et al. 2004; Yaniv 2004a). Furthermore, high quality advice from solicited, *expert* sources is more likely to be implemented than other types of advice (Jungerman and Fischer 2005; Jungermann 1999; Yaniv and Kleinberger 2000; Yaniv and Milyavsky 2007; Gino 2008; Gibbons et al. 2003). The tax practice comprises a substantial percentage of accounting firm revenue (Zeff 2003; Public Accounting Report 2008), tax planning is a large portion of the services that tax practitioners provide, and accounting firms invest in training and knowledge to develop tax plans for their clients (Solomon et al. 1999; Ferguson et al. 2003). Thus, tax experts possess superior knowledge of industry-specific tax avoidance techniques, as well as general tax planning opportunities (McGuire et al. 2012) and recent innovations in tax planning, all of which can improve their clients' abilities to develop effective tax strategies (Deloitte Development LLC 2013; Ernst & Young 2013; PwC 2013). Because of the complex nature of tax planning, managers may consult tax experts for advice prior to implementing a tax strategy.

Ex ante, expert tax advice could be associated with either sustainability or minimization. The managers of firms with sustainable tax strategies will likely seek information about recurring tax avoidance activities that have been consistently upheld by taxing authorities. It is conceivable that tax experts possess this type of information because practitioners market tax strategies with sustainable outcomes (Strategic Capital Partners, LLC 2012; Deloitte LLP 2013; KPMG LLP 2007; PwC 2008). If expert tax advice improves firms' abilities to implement tax strategies that generate consistent tax outcomes, I expect hiring a tax expert audit firm for tax services is associated with a higher likelihood of adopting a sustainable tax strategy. In contrast, employing a minimization tax strategy requires managers to engage in a variety of tax planning opportunities, increasing the importance of information about tax

law changes, new tax planning techniques, and industry-specific tax avoidance. Tax experts possess this information (Deloitte Development LLC 2013; Ernst & Young 2013; PwC 2013), and given the difficulty and cost to individual companies of developing tax planning ideas (Bonner et al. 1992; Deloitte Development LLC 2013; Ernst & Young 2013; McGuire et al. 2012; SD Mayer & Associates LLP 2013), firms emphasizing minimization may benefit from tax experts' specialized knowledge. Furthermore, McGuire et al. (2012) find that firms that purchase tax services from their tax expert audit firms avoid more taxes on average. If expert tax advice facilitates reducing firms' tax burdens, I expect hiring a tax expert audit firm for tax services to increase the likelihood of adopting a minimization tax strategy.

Expert tax advice could influence either tax strategy choice; thus, I state the following hypothesis:

H2: Hiring a tax expert audit firm for tax services will influence a firm's choice of tax strategy.

Information Transfer among Directors

Relationships among firms' directors form an extensive network across which information can be shared. Social network theory indicates that members' network locations can predict behavior (Borgatti and Halgin 2011; Phelps et al. 2012); thus, directors' interactions may influence strategy choices. For example, prior research finds that both value-increasing (e.g., Gulati and Westphal 1999; Haunschild 1993; Palmer et al. 1986; Palmer et al. 1989) and value-decreasing (e.g., Bizjak et al. 2009; Dooley 1969; Pennings 1980) corporate policies travel across firms through directors' connections. These studies demonstrate that firms are more likely to implement a policy or strategy found at another firm when there are connections among directors. Tax planning information can travel similarly; in fact, prior research finds that tax shelter use spreads across firms via interlocking directors (Brown 2011) and firms with board interlocks pay similar levels of tax (Brown and Drake 2014). Furthermore, tax directors meet with the board regularly to discuss tax planning (ATA 2014), indicating that directors communicate directly with the individuals who implement tax strategy.¹⁰ Thus, anecdotal and empirical evidence suggests that information transferred through directors' connections could affect tax strategy choice.

¹⁰ Directors may lack the technical expertise to explain a tax planning technique, but they can share tax planning information by introducing the directors of other companies to the entity responsible for their firm's tax strategy.

Ex ante, information transferred through directors' connections could be associated with either sustainability or minimization. To achieve consistent tax outcomes, managers invest in recurring tax avoidance activities that are more likely to be sustained upon audit (Deloitte LLP 2013). Many firms exhibit more sustainable tax outcomes (McGuire et al. 2013); thus, the directors of these firms could be sharing information that identifies tax planning opportunities previously upheld by taxing authorities. Furthermore, accounting firms, which directors may have done business with, market sustainable tax strategies (Deloitte LLP 2013; KPMG LLP 2007; PwC 2008). Thus, the tax planning information transferred through directors' connections may increase the likelihood that a firm adopts a sustainable tax strategy. In contrast, firms focusing on minimization employ all available tax avoidance mechanisms, which requires information about new tax planning opportunities and the tax avoidance activities being used by their peers and competitors. Prior research suggests that directors may be informed about mechanisms for tax minimization because Brown (2011) finds that board interlocks are associated with specific transactions (i.e., tax shelters) whose goal is to minimize taxes. Her results imply that directors' connections, and the information obtained through those connections, may be an important determinant of focusing on a minimization tax strategy. Therefore, I expect information transferred through directors' connections could also increase the likelihood that a firm chooses a minimization tax strategy.

Directors have access to different types of tax planning information through their connections to other firms. This information is likely a combination of innovative tax planning opportunities and more predictable tax avoidance activities; thus, information transferred through directors' connections could increase the likelihood of choosing either tax strategy. Therefore, I hypothesize that:

***H3:** The information transferred to a firm through its directors' connections will influence a firm's choice of tax strategy.*

IV. Research Design

Sample Selection

To investigate my hypotheses, I obtain a sample of firm-year observations at the intersection of the Compstat, CRSP, Execucomp, Audit Analytics, and Risk Metrics databases from 2000-2010.

Consistent with prior literature, all firms in the sample are incorporated in the U.S., have total assets greater than zero and positive pre-tax book income in the current year, and do not operate in the financial services or utilities sectors. I also require that the observations have non-missing data for cash taxes paid and pre-tax book income for three of the five years necessary to calculate sustainability. Finally, to maintain a consistent sample, I exclude firm-year observations that lack the data necessary to calculate any of the variables used in the analysis – the tax strategy variables, variables of interest, or controls. My final sample consists of 4,668 firm-year observations (1,137 unique firms).

Description of Variables

Following prior research, I define a sustainable tax strategy as a tax strategy that achieves a consistent tax outcome over time, regardless of the tax outcome's level (e.g., McGuire et al. 2013; Neuman et al. 2013), and a minimization tax strategy as a tax strategy that focuses on obtaining the lowest possible tax outcome (e.g., Dyreng et al. 2008; Scholes et al. 2009). Tax strategies are defined in terms of a goal for long-term tax outcomes; thus, to isolate firms' tax strategies, I observe firms' tax outcomes over time. Specifically, I examine the distribution of tax outcomes and identify firms focusing on one of the two dimensions of tax strategy. My approach of inferring firms' unobservable tax strategies from observable tax outcomes measured over a period of time is consistent with the methodology of empirical analyses of business strategy (e.g., Bentley et al. 2012; Higgins et al. 2013) in which firm characteristics derived from financial statement data are used to operationalize unobservable business strategy.

To identify firms adopting a sustainable tax strategy, I use the coefficient of variation of annual cash ETRs (*CV_CETR*) measured over the five-year period, t to $t-4$ (McGuire et al. 2013). *CV_CETR* is calculated as the standard deviation of annual cash ETRs scaled by the absolute value of the mean of annual cash ETRs and measures the variability in firms' tax outcome levels over the period. Firms with low variability maintain a consistent range of cash ETRs over time; thus, I rank firms by year according to *CV_CETR*.¹¹ I classify a firm as following a sustainable tax strategy, *SUSTAIN* equal to one, if its

¹¹ Due to the ranking procedure, firms classified as adopting a sustainable tax strategy will vary with the underlying population; thus, I define *SUSTAIN* using all available observations with data necessary to calculate *CV_CETR*.

CV_CETR is ranked in the lowest quintile for the year, and zero otherwise.¹² Coca-Cola, Conoco-Phillips, and United Parcel Service are examples of firms in my sample that emphasize sustainability.

To identify firms as following a minimization tax strategy, I use the cash ETR measured over the three-year period, t to $t-2$, following Dyreng et al. (2008).¹³ The three-year cash ETR (*CETR3*) equals the sum of cash taxes paid divided by the sum of pre-tax book income less special items and measures the average tax outcome over the three-year period. Firms with low cash ETRs achieve a relatively high level of tax avoidance over time; thus, a firm is classified as following a minimization tax strategy, *MINIMIZE* equal to one, if its *CETR3* is less than or equal to 20 percent, and zero otherwise.¹⁴ Apple, Eli Lilly, and International Paper are several firms in my sample that emphasize minimization.¹⁵

I measure tax strategies using cash ETRs for several reasons. First, firms' tax strategies are ultimately reflected in the cash paid to taxing authorities. Firms have more discretion over the calculation of financial statement income than taxable income (Hanlon et al. 2005); thus, GAAP ETRs may suggest that a firm has adopted a particular tax strategy, but cash taxes paid are likely more representative of the firm's execution of its tax strategy. GAAP ETRs and cash ETRs also capture different aspects of tax planning. GAAP ETRs do not include tax deferral, but are affected by changes in tax accruals (i.e., the valuation allowance or the tax reserve). In contrast, cash ETRs reflect deferred taxes and are not affected by changes in tax accruals (Hanlon and Heitzman 2010). Because cash ETRs are more comprehensive measures of tax planning (i.e., temporary *and* permanent differences) that are not impacted by the judgment inherent in the tax accrual process, I operationalize firms' tax strategies using cash ETRs.¹⁶

¹² Firms with extremely high non-tax costs may forego tax avoidance and pay tax at the top corporate statutory tax rate. To ensure that my results for sustainable tax strategies are not due to firms that do not engage in tax planning, I restrict my sample to observations with a *CETR3* less than 35 percent and re-estimate my regressions. My conclusions are unchanged (i.e., the sign and significance of all coefficients of interest remains the same).

¹³ I measure minimization over a three-year period because Dechow and Dichev (2002) argue that most firm operations, and the associated cash flows, complete their accounting cycle within three years. I estimate *CV_CETR* over five years to be consistent with prior literature (e.g., Minton and Schrand 1999; Minton et al. 2002). My inferences are unchanged when I use the five-year cash ETR to identify firms that emphasize minimization.

¹⁴ I define the minimization tax strategy as a *CETR3* less than or equal to 20 percent because Dyreng et al. (2008) define low tax rate firms as those with a rate of 20 percent or less (p. 62).

¹⁵ Please see Appendix A for detailed definitions of all variables used in this study.

¹⁶ Cash ETRs are not perfect measures of firms' tax activities (e.g., they are affected by stock option expensing). My inferences are similar if I measure tax strategy using GAAP ETRs (please see Section VI).

Method

The decision to focus on sustainability or minimization cannot be separated because it is difficult to emphasize two dimensions of a strategy (March 1991). Therefore, the probability of adopting one tax strategy must be estimated assuming a joint distribution of both tax strategy choices (Wooldridge 2010, 595-6). To control for the interdependence of this choice, I employ seemingly unrelated bivariate probit regression, which models binary relationships whose probability of occurring influences the probability of another event or decision occurring (Mallar 1977). This methodology controls for the simultaneous nature of the decision to focus on one tax strategy and forego emphasizing the alternative tax strategy, providing consistent coefficient estimates for the effect of the variables of interest on tax strategy choice.

To examine whether managerial incentives and practices influence the choice of tax strategy, I estimate the following systems of equations:

$$SUSTAIN_{it} = \alpha + \beta_1 INCENTIVES/PRACTICES_{it} + \beta_{2-7} TAX PLANNING_{it} + \beta_{8-10} PERFORM_{it} + \beta_{11-12} INVEST/REPORT_{it} + \beta_{13} IMR_{it} + industry + \varepsilon \quad (1)$$

$$MINIMIZE_{it} = \alpha + \beta_1 INCENTIVES/PRACTICES_{it} + \beta_{2-7} TAX PLANNING_{it} + \beta_{8-10} PERFORM_{it} + \beta_{11-12} INVEST/REPORT_{it} + \beta_{13} IMR_{it} + industry + \varepsilon, \quad (2)$$

where *SUSTAIN* is an indicator variable equal to one if the firm focuses primarily on a sustainable tax strategy, and zero otherwise, and *MINIMIZE* is an indicator variable equal to one if the firm focuses primarily on a minimization tax strategy, and zero otherwise (both defined above).

INCENTIVES/PRACTICES represents one of the four variables of interest: *DELTA* and *VEGA*, *TAXEXPERT*, and *INFO TRANSFER*, to be defined below. All variables are defined in Appendix A.

I also include controls for firm characteristics, performance, and investing and reporting decisions that could affect firms' tax strategy choices.¹⁷ First, I control for investments in tax planning, *TAX PLANNING*, which represent firms' available tax planning opportunities (Mills et al. 1998). I proxy for

¹⁷ To develop a parsimonious model, I follow the procedures outlined by Hosmer and Lemeshow to identify a comprehensive, yet concise, set of control variables (Agresti 2007); however, I also examine the robustness of my results (discussed in Section V) to the inclusion of additional control variables.

firm size using the natural logarithm of sales (*SIZE*) and leverage using total long-term debt scaled by end-of-the-year total assets (*LEVERAGE*). I control for property, plant, and equipment, net of depreciation, scaled by end-of-year total assets (*PPE*); foreign operations, an indicator variable equal to one if the firm has foreign income and zero otherwise (*FOR_OP*); tax loss carryforwards, an indicator variable equal to one if the firm reports a tax loss carryforward and zero otherwise (*TLCF*); and complexity, the number of segments a company reports in a given year (*SEGMENTS*).

I also control for firm performance which will influence the choice of tax strategy because a firm cannot owe taxes without earning a profit; resources for tax planning (income and cash flow) are derived from performance; and performance volatility represents a non-tax cost. I measure a firm's pre-tax earnings volatility using the coefficient of variation of pre-tax book income (*CV_PTBI*), the standard deviation of annual pre-tax book income scaled by the absolute value of the mean of pre-tax book income over the five-year period, $t-4$ to t . In addition, I control for the firm's profitability using return-on-assets (*ROA*) and the book-to-market ratio (*BTM*). I also include two measures of investing and reporting decisions because these policies are likely governed by the same guiding principles as tax strategy and investment policy can have significant tax consequences. I proxy for a firm's investment in innovative products using research and development expense (*RD*) and control for a firm's financial reporting decisions using the signed value of performance-adjusted discretionary accruals (*DISC_ACC*), following Kothari et al. (2005). Further, when *TAXEXPERT* represents *INCENTIVES/PRACTICES*, I also include *IMR*, the inverse Mills ratio estimated from the selection model to be presented below, to control for the decision to purchase tax services from the audit firm. Finally, I include industry (two-digit SIC code) fixed effects to control for different environments and opportunities across industries.¹⁸

Hypothesis One – Equity Incentives

To investigate whether managerial incentives are associated with firms' decisions to focus on a sustainable or minimization tax strategy, I simultaneously estimate models (1) and (2) including proxies

¹⁸ My results are also robust to including proxies for firms' business strategies (Bentley et al. 2012) and life cycles (Dickinson 2011) (i.e., all coefficients of interest maintain the same sign and level of significance).

for managers' equity incentives – *DELTA*, the sensitivity of the CEO's wealth to changes in the stock price, and *VEGA*, the sensitivity of the CEO's wealth to stock return volatility.¹⁹ Following Core and Guay (1999), I calculate *DELTA* as the change in the CEO's wealth for a one percent change in the firm's stock price. I measure *VEGA* as the change in the value of the CEO's stock option portfolio for a one percent change in stock return volatility (Guay 1999). A positive (negative) and significant coefficient on *DELTA* or *VEGA* is consistent with firms being more (less) likely to concentrate on either the sustainability or minimization dimension of tax strategy when managers have higher values of *DELTA* or *VEGA*. As stated in Hypothesis 1, I expect the coefficient on *DELTA* to be positive for firms focusing on sustainability and the coefficient on *VEGA* to be positive for firms emphasizing minimization.

Hypothesis Two – Expert Tax Advice

To examine whether expert tax advice influences tax strategy choice, I simultaneously estimate models (1) and (2) using audit firm tax expertise to proxy for the advice managers obtain from tax experts.²⁰ My variable of interest is *TAXEXPERT*, an indicator variable equal to one if the firm hires its tax expert audit firm for tax services. Following McGuire et al. (2012), I define tax expertise as a tax service market share of 30 percent (25 percent) or more in years 2003 and later (2002 and earlier) within a Metropolitan Statistical Area (MSA) and industry, and zero otherwise. A positive (negative) and significant coefficient on *TAXEXPERT* is consistent with firms with access to expert tax advice being more (less) likely to concentrate on either a sustainable or minimization tax strategy. Hypothesis 2 suggests that the coefficient on *TAXEXPERT* will be positive for firms engaging in either tax strategy.

The decision to hire the audit firm for tax services is not random; it is a function of observable and unobservable factors (Lassila et al. 2010; McGuire et al. 2012). If the factors that influence firms' decisions to hire the audit firm for tax services also affect the association between expert tax advice and

¹⁹ Following prior research (e.g., Chava and Purnanandam 2010; Coles et al. 2006; Rego and Wilson 2012), I simultaneously estimate the effect of *DELTA* and *VEGA* on the choice of tax strategy to control for both the overlapping and competing nature of these incentives within the executive's compensation contract.

²⁰ A firm could also obtain professional advice from a tax expert that is not its auditor; however, publicly disclosed information on tax service provision is limited to firms that hire their auditor for tax services. Thus, due to current data availability, audit firm tax expertise is the best publicly available proxy for expert tax advice.

tax strategy choice, my analysis of Hypothesis 2 is subject to selection bias. Consistent with prior research in accounting (e.g., Badertscher et al. 2009; McGuire et al. 2012; Omer et al. 2006; Tucker 2007), I employ the Heckman (1979) two-stage approach to control for potential selection bias. Specifically, following McGuire et al. (2012), I estimate a probit regression that approximates the likelihood that a client hires its audit firm for tax services:

$$\begin{aligned}
 TAXSERVICE_{it} = & \alpha + \beta_1 TENURE_{it} + \beta_2 INSTOWN_{it-1} + \beta_3 MERGER_{it} + \beta_4 AUDIND_{it} \\
 & + \beta_5 LNAUDITFEES_{it} + \beta_6 OPPORTUNITY_{it} + \beta_7 SIZE_{it} + \beta_8 DISC_ACC_{it} \\
 & + \beta_9 TLCF_{it} + \beta_{10} \Delta TLCF_{it} + \beta_{11} EQINC_{it} + \beta_{12} FOR_OP_{it} + \beta_{13} RD_{it} \\
 & + \beta_{14} LEVERAGE_{it} + \beta_{15} BTM_{it} + \beta_{16} PPE_{it} + \beta_{17} ROA_{it} + \beta_{18} CASH_{it} \\
 & + \beta_{19} DEP_{it} + \beta_{20} BIG4_{it} + \beta_{21} SECTIER_{it} + industry + year + \varepsilon,
 \end{aligned} \tag{3}$$

where all variables are defined in Appendix A. Using the coefficient estimates from model (3), I calculate the inverse Mills ratio, *IMR*, and include it as a control variable in my second-stage regressions (i.e., models (1) and (2)). The inverse Mills ratio corrects for selection bias by controlling for observable and unobservable factors that influence firms' decisions to hire the audit firm for tax services and also potentially affect the association between expert tax advice and tax strategy choice (Heckman 1979). I present the results of model (3) in Table 4.

Hypothesis Three – Information Transfer among Directors

Finally, I examine whether information transferred through directors' connections influences firms' tax strategy choices by simultaneously estimating models (1) and (2) including *INFO TRANSFER*. *INFO TRANSFER* is a vector of four director centrality proxies, which estimate access to and transfer of information from other firms, measured using principal components factor analysis. In social network analysis, measures of centrality are used to identify the importance of connections within the network and to study interactions and information transfer among individuals (Bonacich 1987; Freeman 1979; Newman 2010). The four centrality proxies that I use are *NUM_CONNECTIONS*, *CONTROL*, *SPEED*, and *CONNECTION_STRENGTH*. *NUM_CONNECTIONS* is calculated as the number of unique connections a director has with other directors in the network and proxies for access to information;

directors with more connections have better access to information. *CONTROL* is calculated as the number of paths between two directors in the network, *a* and *b*, that pass through director *i* divided by the total number of paths between directors *a* and *b* and represents the control that director *i* has over the flow of information from one area of the network to another. Directors with higher values of *CONTROL* are located on more paths between other directors in the network, and thus, control the flow of information.

The third proxy, *SPEED*, is the inverse of the sum of the shortest distances of the paths between the director and other directors in the network and captures the speed at which a director receives information within a network. Directors with higher *SPEED* values receive information more quickly. Finally, *CONNECTION_STRENGTH* is the number of unique connections a director has with other directors, where each of these connections is weighted based upon how well-connected those directors are within the network. *CONNECTION_STRENGTH* assumes that connections to other well-connected directors are more valuable, and directors with these connections receive a higher score.²¹

I calculate each measure at the director level and aggregate these scores for each board; thus, I examine the impact of information transfer to an entire board on the likelihood of emphasizing either sustainability or minimization.²² In the principal components factor analysis, all four variables load on a single factor with an eigenvalue of 2.78, and the proportion of the variance explained by the factor is approximately 70 percent.²³ A positive (negative) and significant coefficient on *INFO TRANSFER* suggests that firms are more (less) likely to focus on either sustainability or minimization when directors have greater access to information from other firms' directors. Hypothesis 3 suggests that the coefficient on *INFO TRANSFER* will be positive for firms focusing on either sustainability or minimization.

²¹ The social network analysis literature refers to *NUM_CONNECTIONS* as degree, *CONTROL* as betweenness centrality, *SPEED* as closeness centrality, and *CONNECTION_STRENGTH* as eigenvector centrality (Newman 2010). My calculations are consistent with this literature, but I revised the variable names for interpretability.

²² I only include observations connected to the largest component of the network in my sample. This component comprises approximately 83 percent of all directors with data necessary to calculate the centrality measures.

²³ The factor loadings of *NUM_CONNECTIONS*, *CONTROL*, *SPEED*, and *CONNECTION_STRENGTH* are 0.95, 0.86, 0.87, and 0.61, respectively.

Relative Importance

These managerial incentives and practices are unlikely equally important in determining the primary focus of firms' tax strategies. To investigate the relative importance of these factors for tax strategy choice, I simultaneously include all variables of interest in the analysis and compare the fully standardized coefficients for these variables in each tax strategy model. I do not make predictions about the relative importance of the managerial incentives and practices for the choice of tax strategy because it is unclear *ex ante* which of these factors will be the most important determinant of each tax strategy.

I use contemporaneous values of managerial incentives and practices to examine the determinants of firms' tax strategy choices for several reasons. First, prior research suggests that nonrecurring tax avoidance activities, which are identified annually, are commonly used (e.g., Mills et al. 1998). Also, advice and information are only valuable when relevant. Some information or advice may be applicable for years, but timeliness affects relevance, especially in a rapidly changing environment. Thus, given the continuous search for new tax planning ideas and the changing tax environment, the current year's information will be important for tax strategy choice. Finally, managers' equity incentives accumulate over time and firms rebalance managers' incentives annually to reflect current objectives; thus, decision-making in the current period, including tax strategy choice, will be a function of this year's incentives.²⁴

V. Results

Descriptive Statistics

Table 1, Panel A provides descriptive statistics for the tax strategy measures, managerial incentives and practices, and the controls for investments in tax planning, firm performance, and investing and reporting choices for the full sample. Panels B, C, and D of Table 1 present descriptive statistics for the two tax strategy subsamples, as well as the sample of firms that are not classified as engaging in either sustainable or minimization tax strategies (i.e., mixed tax strategies). Univariate tests of mean and median differences in firm characteristics (untabulated) indicate that firms with sustainable and

²⁴ My inferences remain unchanged if I use estimates of managerial incentives and practices from the prior year.

minimization tax strategies are significantly different. Relative to firms focusing on minimization, firms that employ sustainable tax strategies are larger and more segmented, but less levered and less likely to report a tax loss carryforward (all $p \leq 0.01$). Firms emphasizing sustainability also have lower earnings volatility and book-to-market ratios and invest less in research and development, but earn a higher return-on-assets than firms emphasizing minimization (all $p < 0.01$). Further, an untabulated analysis of the sample composition reveals that tax strategies are *not* concentrated within particular industries; I find that both tax strategies are present in every two-digit SIC code industry.

Table 2 presents univariate correlations.²⁵ Consistent with my definitions of tax strategy, I find that *CV_CETR* (*CETR3*) is negatively associated with *SUSTAIN* (*MINIMIZE*). In addition, all four proxies of managerial incentives and practices are positively associated with *SUSTAIN*, while *VEGA* is positively associated with *MINIMIZE*. With the exception of the correlation between *TAXEXPERT* and *VEGA*, the variables of interest are positively associated with one another, indicating that they may be used jointly to promote a desired tax strategy. Furthermore, the incentives and practices and tax strategy variables are correlated with many of the control variables, indicating that a multivariate analysis is appropriate to control for the effect of these characteristics on the choice of tax strategy.

Hypothesis One – Equity Incentives

Table 3 presents the results of simultaneously estimating models (1) and (2) to examine the effect of equity incentives on the likelihood of adopting a sustainable or minimization tax strategy. The likelihood ratio test of independence indicates that the two tax strategy models are not independent ($p < 0.01$); thus, simultaneous estimation is appropriate. I find a positive and significant coefficient on *DELTA* ($p < 0.01$) in column (1), indicating that firms with CEOs whose wealth is more sensitive to changes in the stock price are more likely to focus on sustainability. This result reflects the fact that consistently achieving a tax outcome improves the firm's ability to meet earnings' targets, resulting in a stable and positively growing stock price. The average marginal effect in column (3) indicates that a \$1,000,000

²⁵ All coefficients discussed here (and in bold in Table 2) are significant at a minimum of the five percent level.

increase in the CEO's wealth increases the probability that the firm chooses a sustainable tax strategy by 0.009.²⁶ In column (2), the positive and significant coefficient on *VEGA* ($p < 0.01$) suggests that the firm is more likely to emphasize minimization as the sensitivity of the CEO's wealth to stock return volatility increases, consistent with the CEO being rewarded for the increased volatility of net income and returns associated with riskier tax planning activities and their wider range of possible outcomes.²⁷ The average marginal effects reported in columns (3) and (4) indicate that a \$1,000,000 increase in the CEO's stock option portfolio increases (decreases) the probability that a firm focuses on minimization (sustainability) by 0.097 (0.053).²⁸ Collectively, my results support Hypothesis 1 and suggest that managers' tax strategy choices reflect the equity incentives provided by their compensation contracts.²⁹

Hypothesis Two – Expert Tax Advice

Table 4 presents the results of model (3), which estimates the probability that the firm will hire their audit firm for tax services. The selection model exhibits good discriminatory power with an area under the ROC curve of 0.83 (Hosmer and Lemeshow 2002). My results are generally consistent with those of McGuire et al. (2012) and Lassila et al. (2010). Notably, audit firm tenure and audit fees are associated with a higher probability that the client will hire the audit firm for tax services, while the magnitude of non-tax, non-audit fees is negatively associated with the probability that the client will hire the audit firm for tax services. The inverse Mills ratio, *IMR*, is calculated from the coefficient estimates reported in Table 4 and included as an additional control variable in models (1) and (2).

²⁶ The average total compensation for S&P 500 CEOs was \$12.3 million in 2012, an increase of 8 percent over the previous year (AFL-CIO 2013; Popelka 2013), which corresponds to an increase of approximately \$1,000,000.

²⁷ My inferences are unchanged when I also control for the volatility of return-on-assets and prior tax avoidance (i.e., three-year cash ETR measured from $t-2$ to $t-4$).

²⁸ I also examine whether equity incentives interact with either expert tax advice or information from directors' connections to influence the choice of tax strategy by including *DELTA* and *VEGA* and their interaction terms with *TAXEXPERT* and *INFO TRANSFER*, respectively, in the analysis of Hypotheses 2 and 3. I do not find a consistent interaction effect, and my inferences with respect to the main effects of all variables of interest remain the same.

²⁹ Given the CEO's influence over the firm (e.g., Dyreng et al. 2010; Bertrand and Schoar 2003; Bamber et al. 2010), I examined the sensitivity of firms' tax strategy choices to changes in CEO. My results (untabulated) indicate that CEO changes are not systematically related to one tax strategy and the average change in *SUSTAIN* or *MINIMIZE* for firms with and without CEO changes is not statistically significant; thus, on average, CEO changes do not impact firms' tax strategy choices.

Table 5 presents the results from simultaneously estimating models (1) and (2) to examine the effect of audit firm tax expertise on the likelihood of emphasizing sustainability or minimization. Simultaneous estimation of tax strategy choice is necessary because the likelihood ratio test of the independence of the two models is significant ($p < 0.01$). Confirming Hypothesis 2, I find a positive and significant coefficient on *TAXEXPERT* ($p < 0.05$) in column (2) which indicates that firms with access to expert tax advice have a higher likelihood of focusing on minimization, suggesting that the expertise of tax practitioners (i.e., knowledge of the tax law, the client's industry, and revenue authorities' procedures) improves a firm's ability to achieve the lowest possible tax outcome. In column (4), the average marginal effect for *TAXEXPERT* suggests that hiring a tax expert audit firm for tax services increases the probability of choosing a minimization tax strategy by 0.033.³⁰ The negative, but insignificant coefficient on *TAXEXPERT* in column (1) implies that expert tax advice is not a significant determinant of the decision to emphasize a sustainable tax strategy perhaps because tax experts offer more information about new or industry-specific tax planning techniques than more predictable tax planning opportunities.^{31, 32}

Hypothesis Three – Information Transfer among Directors

Table 6 presents the results of simultaneously estimating models (1) and (2) to examine the effect of information transferred through directors' connections on the likelihood that firms emphasize a sustainable or minimization tax strategy. The two choices of tax strategy are not independent ($p < 0.01$); thus, simultaneous estimation is appropriate. Consistent with Hypothesis 3, information transferred through directors' connections is an important determinant of tax strategy choice. Specifically, in column (1), I find a negative and significant coefficient on *INFO TRANSFER* ($p < 0.05$), indicating that firms

³⁰ The marginal effects presented in Table 5 are estimated for the regression model excluding *IMR* because the post-estimation procedures for bivariate probit regression cannot compute marginal effects for predicted variables.

³¹ My results are robust to including these additional controls - the change in value of the tax loss carryforward; the amount of equity income, cash, and depreciation and amortization; the market value of the company relative to the value of all possible clients in the same MSA and industry; Big 4 or second-tier (Grant Thornton or BDO Seidman) audit firms; and prior tax avoidance (i.e., three-year cash ETR measured from $t-2$ to $t-4$) (McGuire et al. 2012).

³² In an untabulated analysis, I examine the sensitivity of my results to the level of tax fees paid to the auditor. I separately include indicator variables for tax fees scaled by audit fees greater than the mean, median, or third quartile and the interaction with *TAXEXPERT*. In all instances, I find that neither the tax fee indicator, nor the interaction between the tax fee indicator and *TAXEXPERT* are significant, which suggests that my results are not solely due to firms paying high tax fees to their auditor, an indicator of tax planning (e.g., Dhaliwal et al. 2013).

whose directors have more connections to other directors, greater access to information, and more control over the flow of information have a lower likelihood of focusing on sustainability. In contrast, the positive and significant coefficient on *INFO TRANSFER* ($p < 0.01$) in column (2) suggests that information transferred through directors' connections is associated with a higher likelihood of focusing on minimization.³³ The average marginal effects reported in columns (3) and (4) indicate that a one unit increase in *INFO TRANSFER* (i.e., an increase in director connectedness) increases (decreases) the probability of emphasizing minimization (sustainability) by 0.021 (0.019). Collectively, these results suggest that firms with greater access to information through well-connected directors are less likely to focus on identifying opportunities to maintain a consistent tax outcome (i.e., to choose a sustainable tax strategy) and are more likely to leverage the information to reduce taxes and emphasize minimization.

Relative Importance

Table 7 presents the results of examining the relative importance of managerial incentives and practices for firms' tax strategy choices.^{34,35} The two models are not independent ($p < 0.01$); thus, simultaneous estimation is appropriate. My results are consistent with the separate tests of each factor reported above (i.e., all coefficients retain the same sign and are significant).³⁶ To determine which factor has the most influence on tax strategy choice, I compare the relative size of the fully standardized coefficients reported in Table 7. In column (1), the coefficient on *DELTA (INFO TRANSFER)* is positive (negative) and significant ($p < 0.01$ and $p < 0.05$, respectively). Tests of the difference in the coefficients suggest that the sensitivity of the CEO's wealth to stock price (*DELTA*) is relatively more important than information from directors' connections (*INFO TRANSFER*) to the decision to emphasize sustainability

³³ My results are robust to the inclusion of controls for board characteristics – size, busy directors, and independent directors, as well as prior tax avoidance (i.e., three-year cash ETR measured from $t-2$ to $t-4$).

³⁴ Including all variables in the model does not pose a multicollinearity problem. The highest condition index is 28.5, with the relation between *SIZE* and the intercept contributing significantly to this condition index; however, it is still below the rule of thumb of 30 (Belsley et al. 1980). The second highest condition index is only 11.07.

³⁵ My tax strategy choice models perform well. I conduct classification tests by obtaining the predicted probability that firms engage in particular tax strategies from the analysis presented in Table 7. Using a cutoff of 0.6, I find that the model correctly classifies firms following a sustainable (minimization) tax strategy 83 (75) percent of the time.

³⁶ *INFO TRANSFER* has a lower level of significance in column (2). This coefficient was significant at the one percent level in Table 6, but is now only marginally significant. However, my overall conclusions are unchanged.

($Z = 2.16, p < 0.05$).³⁷ In column (2), the coefficients on *VEGA*, *TAXEXPERT*, and *INFO TRANSFER* are all positive and significant (all $p < 0.10$, at least), but tests of the equality of the coefficients reveal that firms that focus on minimization are influenced most by equity incentives (*VEGA*) ($p < 0.01$ and 0.05), followed by expert tax advice ($p < 0.10$), and finally, information from directors' connections.³⁸

The results of my analyses suggest that, among my variables of interest, equity incentives are the most important determinants of tax strategy choice, followed by obtaining tax planning information from tax experts or directors at other firms, consistent with managerial practices having a second-order effect on tax strategy choice. My results demonstrate that equity incentives are not only associated with tax outcomes, as provided in the prior literature, but are the primary determinants of tax strategy choice, which produces observed tax outcomes. Furthermore, managerial incentives outweigh the effects of management practices on tax strategy choice.

Matched Firm Analysis

The income and expenses of different activities (e.g., acquiring a new asset versus operating in a foreign country) are not always treated similarly for tax purposes; thus, the tax benefits of many activities are not equal (Inger 2014), resulting in different tax planning opportunities (Mills et al. 1998). If firms emphasizing sustainability have different tax planning opportunities than firms focusing on minimization, it could generate differences in managerial incentives and practices for the two sets of firms that explain my findings. Tax planning opportunities are related to firm characteristics; thus, I also conduct my analysis using a matched sample to hold tax planning opportunities constant while examining the influence of managerial incentives and practices on tax strategy choice. Specifically, I match firms with sustainable tax strategies (*SUSTAIN* = 1) to firms with minimization tax strategies (*MINIMIZE* = 1)

³⁷ I test the coefficients' equality using Paternoster et al.'s (1998, 862) equation (4). Specifically, Z is equal to the difference in the coefficients divided by the square root of the sum of the squared standard errors of the coefficients.

³⁸ The coefficient on *VEGA* is significantly larger than the coefficients on *TAXEXPERT* and *INFO TRANSFER* ($Z = 2.48$ and $-1.72, p < 0.01$ and $p < 0.05$, respectively); thus, equity incentives influence the choice of a minimization tax strategy to a greater extent than expert tax advice or information from directors' connections. The coefficient on *TAXEXPERT* is also significantly larger than the coefficient on *INFO TRANSFER* ($Z = 1.29, p < 0.10$), suggesting that expert tax advice influences an emphasis on minimization more than information from directors' connections.

without replacement on *SIZE* within both industry (Fama-French 17 classification) and year, yielding a sample of 1,998 firm-year observations (999 unique matched pairs). I then re-examine my hypotheses using a logistic regression where the dependent variable is *SUSTAIN* and the independent variables are as previously defined; I present the results in Table 8.

The coefficients on *SIZE* in Table 8 are no longer significant, indicating that the matching procedure identified good pairs; however, to control for residual bias due to imperfect matching, I include *SIZE* and industry fixed effects in the models.³⁹ Similar to my primary results, I find a positive and significant coefficient on *DELTA* ($p < 0.01$), suggesting that firms whose CEO's wealth is more sensitive to changes in the stock price are more likely to implement a sustainable tax strategy than a minimization tax strategy. I also find negative and significant coefficients on *TAXEXPERT* and *INFO TRANSFER* (both $p < 0.05$), indicating that firms with access to external tax planning information are less (more) likely to emphasize sustainability (minimization). Thus, I find that tax planning opportunities are not the only determinant of tax strategy choice, and importantly, differences in opportunities do not subsume the association between managerial incentives and practices and tax strategy choice. Holding these opportunities constant, managerial incentives and practices continue to be significant determinants of firms' tax strategy choices.

VI. Additional Analysis

Firm Value Analysis

According to the strategy literature, there are multiple methods for achieving any strategy's goal (Hambrick 1983; Miller 1987; Porter 2004; Zahra and Covin 1993); thus, I implicitly assumed that both sustainability and minimization create value for firms. I test this assumption by conducting univariate tests of differences in mean and median values of market value of equity (*MVE*) for firms following a sustainable or minimization tax strategy relative to firms with a mixed tax strategy.⁴⁰ I find that firms

³⁹ I also matched firms within industry and year on *SIZE* and *LEVERAGE* and *SIZE*, *LEVERAGE*, and *TLCF*, respectively. These samples yield that same inferences as those reported in Table 8.

⁴⁰ *MVE* is calculated as the product of the number of common shares outstanding and the stock price measured on the last day of the fiscal year, scaled by end-of-year total assets.

focusing on either sustainability or minimization have significantly higher *MVE* than firms following a mixed tax strategy in the current and three subsequent years (all $p < 0.01$). These results indicate that firms explicitly choosing a tax strategy have higher market values than firms with tax strategies that are not well defined, supporting the literature's argument that any dimension of a strategy can create value.

Forward Tax Strategy Analysis

I investigate the association between tax strategy choice and contemporaneous values of managerial incentives and practices to capture the timeliness and relevance of new information received during the current year, as well as the accumulation and rebalancing of equity incentives on an annual basis. However, strategy decisions are sticky (e.g., Bentley et al. 2012), and, although firms can and do change their tax strategies, a firm's tax strategy choice is relatively stationary over time.^{41,42} Thus, to examine the extent to which current managerial incentives and practices influence future decisions, I also investigate firms' future tax strategy choices by substituting proxies for firms' tax strategies measured at a future date, one-, two-, and four-years ahead,⁴³ for current period tax strategy measures.

When the choice of tax strategy is measured in year $t+1$, I find that *VEGA* and *INFO TRANSFER* are negatively associated with sustainability ($p < 0.01$ and $p < 0.10$, respectively) and *VEGA* and *TAXEXPERT* are positively associated with minimization ($p < 0.01$ and $p < 0.05$, respectively). Measuring tax strategy in either year $t+2$ or year $t+4$, I find positive and significant coefficients on *DELTA* and negative and significant coefficients on *TAXEXPERT* for *SUSTAIN* (all $p < 0.10$) and positive and significant coefficients on *VEGA* for *MINIMIZE* (both $p < 0.01$). Thus, current period managerial incentives and practices are associated with future tax strategy choices, implying that these factors influence one-time tax planning activities and changes to tax planning, as well as recurring tax avoidance

⁴¹ Approximately 64 (75) percent of firms in my sample are classified as following a sustainable (minimization) tax strategy for 50 percent of their firm-year observations.

⁴² No firms switch between tax strategies, but an examination of year-over-year changes reveals 82 observations that shift more than one quintile of *CV_CETR* and 260 observations that experience a change in *CETR3* of more than 10 percent. As a robustness test, I control for these 342 potentially influential observations and find that my inferences are unchanged (i.e., all coefficients of interest retain the same sign and level of significance).

⁴³ I measure tax strategy four years into the future because five years of data is required to measure *CV_CETR*. Thus, in year $t+4$, *CV_CETR* is comprised of cash ETRs from years $t+4$ to t , only current and future tax outcomes.

activities. Furthermore, as tax strategy is measured further out in time, the impact of these factors attenuates in a manner consistent with the relative importance analysis. Specifically, I find that managerial incentives continue to be highly significant predictors of the choice of tax strategy four-years ahead, but director connectedness is no longer significantly associated with tax strategy choice and the significance of expert tax advice is lessened. This attenuation parallels my findings in Table 7 – within these factors, managerial incentives are the most important determinants of tax strategy choice followed by obtaining expert tax advice and information from directors' connections.

Tax Strategy Classification using GAAP ETR

I define tax strategies using cash ETRs because firms' tax strategies ultimately manifest in cash outflow to taxing authorities, firms have less discretion in the calculation of taxable income than financial statement income (Hanlon et al. 2005), and firms often manage earnings to meet benchmarks or present performance more favorably (e.g., Dechow et al. 2010). However, prior literature also provides evidence that many firms are concerned about the financial statement ETR, or GAAP ETR (Phillips 2003; Robinson et al. 2010; Armstrong et al. 2012; Powers et al. 2013); thus, I extend my main analysis by identifying tax strategy using GAAP ETRs. Here, I classify a firm as following a sustainable tax strategy if its coefficient of variation of annual GAAP ETRs (CV_GETR) measured over the five-year period, t to $t-4$, is in the lowest quintile for the year and as following a minimization tax strategy if its three-year GAAP ETR ($GETR3$) measured over the period t to $t-2$ is less than or equal to 20 percent. I re-estimate models (1) and (2) defining firms as following sustainable or minimization tax strategies if both cash and GAAP ETRs indicate the same tax strategy choice.⁴⁴ My inferences regarding the effect of managerial practices on firms' tax strategy choices are unchanged when tax strategy is measured using GAAP ETRs. Specifically, I find that *TAXEXPERT* ($p < 0.05$) and *INFO TRANSFER* ($p < 0.10$) are positively associated with minimization, while *INFO TRANSFER* is negatively associated with sustainability ($p <$

⁴⁴ I define tax strategy in this manner because tax expense and cash taxes paid are products of significantly different calculations subject to different levels of judgment. Consistent with these measures representing different aspects of firms' tax burdens, they produce different tax strategy samples (i.e., 51 (35) percent of firms identified as focusing on sustainability (minimization) using cash ETRs are classified as following that tax strategy using GAAP ETRs).

0.01). However, although the positive coefficient on *DELTA* (*VEGA*) implies that firms whose CEOs' wealth is more sensitive to changes in the stock price (stock return volatility) are more likely to adopt a sustainable (minimization) tax strategy, neither coefficient is statistically significant. Therefore, my results regarding managerial incentives are attenuated when I measure tax strategy using GAAP ETRs.

Mixed Tax Strategy Analysis

Following prior studies (e.g., Bentley et al. 2012; Higgins et al. 2013), I focus on identifying and examining firms that choose a salient tax strategy. However, although difficult to implement effectively, firms can have mixed strategies which incorporate *elements* of multiple dimensions of a strategy (Miles et al. 1978; Porter 2004). A sizeable portion of my sample is not classified as adopting either a sustainable or minimization tax strategy; thus, I investigate whether managerial incentives and practices influence the choice of a mixed tax strategy.⁴⁵ Univariate tests indicate that mixed tax strategy firms are smaller, more likely to report a TLMCF, have more volatile pre-tax book income, and engage in more R&D than firms emphasizing sustainability (all $p < 0.01$), but are more capital intensive and segmented, less likely to have foreign operations or report a TLMCF, have less volatile pre-tax book income, and engage in less R&D than firms emphasizing minimization (all $p < 0.05$). I employ multinomial logistic regression to compare the effect of managerial incentives and practices on the likelihood that a firm chooses a mixed tax strategy relative to either sustainability or minimization. My results (untabulated) indicate that firms with mixed tax strategies behave differently than firms with more salient tax strategies. Specifically, the coefficients on the variables of interest for the mixed tax strategy firms are significant in the opposite direction of the coefficients previously reported for *SUSTAIN* and *MINIMIZE*, reiterating the importance of identifying firms that choose a salient tax strategy for examining the determinants of tax strategy choice.

Both Tax Strategy Analysis

Despite the difficulty of emphasizing two dimensions of a strategy at once (March 1991), there are firms that are successful at achieving a multi-dimensional approach to a strategy. Consistent with this

⁴⁵ *MIXED* equals one if both *SUSTAIN* and *MINIMIZE* equal zero, and zero otherwise.

notion, I identify 187 observations that are classified as following both sustainable and minimization tax strategies. Given the small sample, I conduct univariate tests to identify similarities and differences between this sample and firms that focus exclusively on sustainability or minimization. Firms that pursue both dimensions of tax strategy have more volatile cash ETRs than firms that solely emphasize sustainability and higher three-year cash ETRs than firms that primarily focus on minimization (both $p < 0.01$), suggesting that these firms are more moderate, in terms of both the level and variability of tax avoidance, than those that only pursue one dimension of tax strategy. With respect to firm characteristics, these firms are larger and more segmented with higher signed values of discretionary accruals than firms focusing on either sustainability or minimization (all $p < 0.01$); however, relative to firms with sustainable tax strategies, these firms engage in more R&D ($p < 0.01$), but do not exhibit different firm performance. Relative to firms emphasizing minimization, firms that focus on both tax strategies perform better (all $p < 0.01$), but do not differ on the level of investment in R&D. Thus, these firms appear to have more flexibility and more available tax planning opportunities, permitting consistent tax minimization over time. It is also important to note that, on average, relative to *either* firms with sustainable or minimization tax strategies, firms that emphasize both dimensions of tax strategy have higher values of managerial incentives and practices (all $p < 0.10$), implying that these firms not only have the resources and capabilities to focus on both sustainability and minimization, but they also have the necessary incentives and practices in place to encourage managers to focus on both dimensions of tax strategy.

VII. Conclusions

This study examines the managerial incentives and practices associated with firms' decisions to concentrate on sustainable or minimization tax strategies and their relative importance as determinants of firms' tax strategy choices. Sustainability strives to achieve a consistent tax outcome over time, while minimization strives to achieve the lowest possible tax outcome. The goal of effective tax planning, and thus tax strategy, is to maximize after-tax returns. Therefore, understanding the determinants of firms' tax strategy choices is important because tax strategy can contribute significantly to firm value, but researchers know relatively little about why firms choose their tax strategy.

I draw upon prior literature that posits that managerial incentives and practices are significant determinants of managers' decision-making in general (e.g., Jensen and Meckling 1976), and tax strategy, specifically (e.g., Dyreng et al. 2008; Mills et al. 1998). Using a sample of firm-year observations from 2000 to 2010, I find that firms are more likely to focus on a sustainable tax strategy when the CEO's wealth is more sensitive to changes in stock price (i.e., incentivized to increase firm performance) and less likely to emphasize sustainability when the firm has more access to information through its directors' connections. In contrast, firms are more likely to concentrate on a minimization tax strategy when the CEO's wealth is more sensitive to changes in stock return volatility (i.e., incentivized to make riskier decisions), a tax expert audit firm provides the firm's tax services, or firms have more access to information through their directors' connections. Finally, I find that managerial incentives are the most important factors for tax strategy choice followed by the practices of obtaining expert tax advice and information from directors' connections. Unlike prior literature that documents an association between managerial incentives and tax outcomes, I provide evidence that managerial incentives are the most important factor preceding the choice of tax strategy, which produces the observed tax outcomes.

This study contributes to the literature that examines the determinants of corporate tax avoidance by investigating two dimensions of tax strategy and the determinants of the underlying tax strategy (i.e., the decision to emphasize either a sustainable or minimization tax strategy to maximize firm value). Furthermore, although the prior literature has identified many characteristics associated with firms' tax avoidance, much of the variation in tax outcomes has yet to be explained (Hanlon and Heitzman 2010; Shevlin 2007). Tax outcomes depend on firms' tax strategies; thus, by identifying the determinants of tax strategy choice, this study can further explain the variation in observed tax outcomes. In addition, by examining both sustainability and minimization, my study answers Maydew's (2001) call for research that explains why all firms do not minimize taxes. Finally, my findings may be of interest to firms and shareholders because they indicate that managerial incentives and practices can be used to influence the choice of tax strategy; thus, shareholders that desire to alter an existing strategy or implement a new tax strategy may find these mechanisms useful for accomplishing their goal.

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Appendix A Variable Definitions

Variable	Definition
<u>Tax Strategies</u>	
<i>SUSTAIN</i>	= An indicator variable equal to one if the firm-year observation has a coefficient of variation of cash ETRs (<i>CV_CETR</i>) in the lowest quintile for the year and equal to zero otherwise.
<i>MINIMIZE</i>	= An indicator variable equal to one if the firm-year observation has a three-year cash ETR (<i>CETR3</i>) less than or equal to 20 percent and equal to zero otherwise.
<i>MIXED</i>	= An indicator variable equal to one if both <i>SUSTAIN</i> and <i>MINIMIZE</i> equal zero for the firm-year observation and equal to zero otherwise.
<i>CV_CETR</i>	= The standard deviation of annual cash ETRs for the five-year period prior to and including the current year (<i>t</i> to <i>t-4</i>) scaled by the absolute value of the mean of annual cash ETRs over the same five-year period.
<i>CETR3</i>	= The ratio of the sum of cash taxes paid (<i>TXPD</i>) divided by the sum of pre-tax book income less special items (<i>PI-SPI</i>) measured over the three-year period, <i>t</i> to <i>t-2</i> .
<u>Managerial Incentives/Practices</u>	
<i>DELTA</i>	= The change in the CEO's wealth for a one percent change in the firm's stock price, following Core and Guay (1999).
<i>VEGA</i>	= The change in value of the CEO's stock option portfolio for a one percent change in stock return volatility, following Core (1999).
<i>TAXEXPERT</i>	= An indicator variable equal to one if the audit firm has a tax service market share equal to 30 percent (25 percent) or more in years 2003 or later (2002 and earlier) within a Metropolitan Statistical Area (MSA) and industry and equal to zero otherwise, following McGuire et al. (2012).
<i>INFO TRANSFER</i>	= The factor score of a vector of four director centrality (connectivity) measures, <i>NUM_CONNECTIONS</i> , <i>CONTROL</i> , <i>SPEED</i> , and <i>CONNECTION_STRENGTH</i> . <i>NUM_CONNECTIONS</i> is the number of unique connections a director has with other directors in the network. <i>CONTROL</i> is calculated as the number of paths between two individuals in the network, <i>a</i> and <i>b</i> , that pass through individual <i>i</i> divided by the total number of paths between the two individuals, <i>a</i> and <i>b</i> . <i>SPEED</i> is the inverse of the sum of the shortest distances of the paths between the director and the other directors in the network. <i>CONNECTION_STRENGTH</i> is measured as the number of unique connections a director has with other directors, but weights each of these connections based upon how well-connected those directors are within the network.
<u>Tax Planning</u>	
<i>SIZE</i>	= The natural logarithm of sales (<i>SALE</i>) plus one.
<i>LEVERAGE</i>	= The ratio of total long-term debt (<i>DLTT</i>) to end-of-the-year total assets (<i>AT</i>).
<i>PPE</i>	= The ratio of property, plant, and equipment (<i>PPE</i>) to end-of-the-year total assets (<i>AT</i>).
<i>FOR_OP</i>	= An indicator variable equal to one if foreign income (<i>PIFO</i>) is greater than zero and equal to zero otherwise.
<i>TLCF</i>	= An indicator variable equal to one if tax loss carryforward (<i>TLCF</i>) is greater than zero and equal to zero otherwise.
<i>SEGMENTS</i>	= The sum of the number of business (<i>BUSSEG</i>) and operating (<i>OPSEG</i>) segments reported during the year.

(continued on next page)

Appendix A (continued)

Variable	Definition
<u>Performance</u>	
<i>CV_PTBI</i>	= The standard deviation of annual pre-tax book income (<i>PI-SPI</i>) for the five-year period prior to and including the current year (<i>t</i> to <i>t-4</i>) scaled by the absolute value of the mean of annual pre-tax book income over the same five-year period.
<i>ROA</i>	= The ratio of income before extraordinary items (<i>IB</i>) to average total assets (<i>AT</i>).
<i>BTM</i>	= The ratio of total common equity (<i>CEQ</i>) to market value of equity (<i>PRCC_F*CSHO</i>).
<u>Investing/Reporting</u>	
<i>RD</i>	= The ratio of research and development expense (<i>XRD</i>) to beginning-of-the-year total assets (<i>AT</i>).
<i>DISC_ACC</i>	= The signed value of the error term from the regression of total accruals ($(IB_t - OANCF_t)/AT_{t-1}$) on the inverse of beginning-of-the-year total assets ($1/AT_{t-1}$), the change in sales adjusted for the change in receivables ($((SALE_t - SALE_{t-1}) - (RECT_t - RECT_{t-1}))/AT_{t-1}$), current year property, plant, and equipment ($PPENT_t/AT_{t-1}$), and prior year return on assets, all scaled by beginning of the year total assets ($(NI_{t-1}/AT_{t-2})/AT_{t-1}$), calculated using the modified Jones (1991) model as modified by Kothari et al. (2005).
<i>IMR</i>	= The inverse Mills ratio calculated from coefficient estimates from model (3).
<u>Additional Controls for Selection Model</u>	
<i>TAXSERVICE</i>	= An indicator variable equal to one if the client purchased tax services from their external audit firm and equal to zero otherwise.
<i>TENURE</i>	= The length of the audit firm's tenure with the client.
<i>INSTOWN</i>	= The percentage of common shares outstanding owned by institutions at the beginning of the year.
<i>MERGER</i>	= An indicator variable equal to one if the firm engaged in any merger or acquisition activity (<i>AQP</i> not equal to missing) during the year and equal to zero otherwise.
<i>AUDIND</i>	= The auditor's independence from the client calculated as non-audit fees less tax service fees received from the client scaled by total audit fees received from the client.
<i>LNAUDITFEES</i>	= The natural logarithm of audit fees received from the client.
<i>OPPORTUNITY</i>	= The market value of the client scaled by the sum of the market values of all clients in the same Metropolitan Statistical Area (MSA) and industry.
<i>ΔTLCF</i>	= The change in the value of the tax loss carryforward (<i>TLCF</i>) from year <i>t-1</i> to year <i>t</i> scaled by beginning-of-the-year total assets (<i>AT</i>).
<i>EQINC</i>	= The equity income of the client (<i>ESUB</i>) scaled by beginning-of-the-year total assets (<i>AT</i>).
<i>CASH</i>	= The cash and cash equivalents (<i>CHE</i>) scaled by beginning-of-the-year total assets (<i>AT</i>).
<i>DEP</i>	= The depreciation and amortization expense (<i>DP</i>) scaled by beginning-of-the-year total assets (<i>AT</i>).
<i>BIG4</i>	= An indicator variable equal to one if the client is audited by a Big 4 accounting firm and equal to zero otherwise.
<i>SECTIER</i>	= An indicator variable equal to one if the client is audited by either Grant Thornton or BDO Seidman and equal to zero otherwise.

TABLE 1
Descriptive Statistics

Panel A: Descriptive Statistics for Full Sample								
Variable	N	Mean	Std Dev	25%	Median	75%	Minimum	Maximum
<u>Tax Strategies</u>								
SUSTAIN	4668	0.327	0.469	0	0	1	0	1
MINIMIZE	4668	0.356	0.479	0	0	1	0	1
CV_CETR	4668	0.584	0.439	0.251	0.456	0.805	0.063	2.175
CETR3	4668	0.246	0.144	0.156	0.244	0.321	0	1
<u>Managerial Incentives/Practices</u>								
DELTA	4668	654.948	1684.490	56.333	164.265	484.487	0.237	12496.130
VEGA	4668	106.356	236.986	0	0	94.084	0	1374.020
TAXEXPERT	4668	0.568	0.495	0	1	1	0	1
INFO	4668	0.000	1.000	-0.716	-0.239	0.526	-1.593	4.249
TRANSFER								
<u>Tax Planning</u>								
SIZE	4668	7.796	1.444	6.764	7.623	8.694	4.725	11.495
LEVERAGE	4668	0.172	0.140	0.041	0.160	0.265	0	0.578
PPE	4668	0.257	0.200	0.108	0.196	0.349	0.021	0.866
FOR_OP	4668	0.136	0.343	0	0	0	0	1
TLCF	4668	0.472	0.499	0	0	1	0	1
SEGMENTS	4668	3.542	2.400	1	3	5	1	21
<u>Performance</u>								
CV_PTBI	4668	0.562	0.892	0.187	0.329	0.566	0.050	6.850
ROA	4668	0.071	0.061	0.036	0.066	0.102	-0.140	0.275
BTM	4668	0.465	0.299	0.257	0.401	0.597	-0.015	1.679
<u>Investing/Reporting</u>								
RD	4668	0.030	0.047	0	0.006	0.041	0	0.215
DISC_ACC	4668	0.052	0.093	-0.003	0.044	0.106	-0.220	0.327
IMR	4668	0.376	0.109	0.311	0.338	0.388	0.288	0.798

(continued on next page)

TABLE 1 (continued)

Panel B: Descriptive Statistics by Tax Strategy – <i>SUSTAIN</i> (1,526 observations)						Panel C: Descriptive Statistics by Tax Strategy – <i>MIXED</i> (1,667 observations)					Panel D: Descriptive Statistics by Tax Strategy – <i>MINIMIZE</i> (1,662 observations)				
Variable	Mean	Std Dev	25%	Median	75%	Mean	Std Dev	25%	Median	75%	Mean	Std Dev	25%	Median	75%
<u>Tax Strategies</u>															
<i>CV_CETR</i>	0.191	0.072	0.136	0.192	0.248	0.634	0.294	0.402	0.56	0.791	0.855	0.495	0.458	0.753	1.161
<i>CETR3</i>	0.287	0.078	0.235	0.291	0.341	0.334	0.148	0.243	0.292	0.365	0.11	0.063	0.059	0.121	0.164
<u>Managerial Incentives/Practices</u>															
<i>DELTA</i>	872.666	2066.32	74.551	210.362	660.713	488.517	1355.51	43.813	126.24	368.389	674.545	1665.16	61.962	185.237	546.368
<i>VEGA</i>	124.134	262.404	0	0	119.636	81.993	202.712	0	0	65.729	125.171	259.879	0	0	121.258
<i>TAXEXPERT</i>	0.603	0.489	0	1	1	0.537	0.499	0	1	1	0.579	0.494	0	1	1
<i>INFO TRANSFER</i>	0.166	1.025	-0.608	-0.033	0.757	-0.094	0.986	-0.792	-0.294	0.371	-0.021	0.995	-0.737	-0.267	0.522
<u>Tax Planning</u>															
<i>SIZE</i>	8.256	1.432	7.218	8.175	9.205	7.657	1.37	6.647	7.492	8.541	7.599	1.463	6.6	7.468	8.483
<i>LEVERAGE</i>	0.166	0.129	0.056	0.158	0.25	0.173	0.14	0.041	0.16	0.265	0.178	0.148	0.034	0.164	0.275
<i>PPE</i>	0.255	0.179	0.129	0.21	0.339	0.254	0.191	0.109	0.205	0.347	0.263	0.226	0.092	0.181	0.367
<i>FOR_OP</i>	0.138	0.345	0	0	0	0.126	0.332	0	0	0	0.147	0.355	0	0	0
<i>TLCF</i>	0.407	0.491	0	0	1	0.481	0.5	0	0	1	0.522	0.5	0	1	1
<i>SEGMENTS</i>	3.649	2.333	1	3	5	3.631	2.531	1	3	5	3.438	2.337	1	3	5
<u>Performance</u>															
<i>CV_PTBI</i>	0.244	0.15	0.138	0.206	0.319	0.562	0.759	0.227	0.383	0.602	0.819	1.222	0.24	0.428	0.781
<i>ROA</i>	0.093	0.054	0.058	0.089	0.122	0.059	0.061	0.028	0.055	0.088	0.065	0.062	0.031	0.058	0.093
<i>BTM</i>	0.386	0.245	0.217	0.336	0.506	0.523	0.321	0.31	0.455	0.661	0.467	0.302	0.252	0.405	0.609
<u>Investing/Reporting</u>															
<i>RD</i>	0.019	0.032	0	0.001	0.025	0.027	0.045	0	0.002	0.036	0.044	0.056	0	0.019	0.074
<i>DISC_ACC</i>	0.055	0.089	0.002	0.044	0.104	0.051	0.095	-0.007	0.046	0.111	0.054	0.095	0.001	0.045	0.108
<i>IMR</i>	0.364	0.101	0.307	0.330	0.373	0.385	0.113	0.316	0.345	0.397	0.374	0.109	0.309	0.334	0.388

Refer to Appendix A for variable definitions.

All variables except *SUSTAIN*, *MINIMIZE*, *MIXED*, *CETR3*, *TAXEXPERT*, *FOR_OP*, *TLCF*, *SEGMENTS*, and *IMR* have been winsorized at the 1st and 99th percentiles.

TABLE 2
Correlations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
<u>Tax Strategies</u>																				
(1) <i>SUSTAIN</i>	1																			
(2) <i>MINIMIZE</i>	-0.340	1																		
(3) <i>CV_CETR</i>	-0.624	0.460	1																	
(4) <i>CETR3</i>	0.197	-0.703	-0.345	1																
<u>Managerial Incentives/Practices</u>																				
(5) <i>DELTA</i>	0.090	0.009	-0.078	-0.009	1															
(6) <i>VEGA</i>	0.052	0.059	-0.031	-0.045	0.255	1														
(7) <i>TAXEXPERT</i>	0.049	0.016	-0.041	-0.002	0.036	-0.041	1													
(8) <i>INFO TRANSFER</i>	0.116	-0.016	-0.126	0.014	0.110	0.398	0.052	1												
<u>Tax Planning</u>																				
(9) <i>SIZE</i>	0.222	-0.101	-0.281	0.073	0.230	0.275	0.175	0.545	1											
(10) <i>LEVERAGE</i>	-0.033	0.029	0.034	-0.047	-0.036	0.032	0.006	0.162	0.165	1										
(11) <i>PPE</i>	-0.006	0.022	0.023	-0.017	0.005	0.020	-0.039	0.073	0.105	0.245	1									
(12) <i>FOR_OP</i>	0.005	0.025	-0.028	-0.024	0.046	0.177	-0.128	0.098	0.055	0.063	-0.013	1								
(13) <i>TLCF</i>	-0.091	0.074	0.095	-0.091	-0.043	-0.064	0.044	-0.046	0.00	0.070	-0.137	0.00	1							
(14) <i>SEGMENTS</i>	0.031	-0.032	-0.064	0.045	0.034	0.095	0.059	0.272	0.289	0.131	-0.027	0.120	0.093	1						
<u>Performance</u>																				
(15) <i>CV_PTBI</i>	-0.249	0.214	0.508	-0.132	-0.026	-0.032	0.003	-0.123	-0.216	-0.024	0.003	-0.023	0.053	-0.056	1					
(16) <i>ROA</i>	0.250	-0.074	-0.239	-0.023	0.092	0.080	0.027	0.022	0.036	-0.255	-0.027	-0.025	-0.122	-0.081	-0.097	1				
(17) <i>BTM</i>	-0.184	0.005	0.163	0.061	-0.145	-0.197	-0.068	-0.179	-0.091	0.029	0.075	-0.034	0.055	0.024	0.087	-0.490	1			
<u>Investing/Reporting</u>																				
(18) <i>RD</i>	-0.162	0.218	0.212	-0.193	0.033	0.076	-0.018	-0.093	-0.221	-0.254	-0.312	-0.011	0.073	-0.120	0.181	0.121	-0.206	1		
(19) <i>DISC_ACC</i>	0.020	0.011	-0.022	0.006	-0.021	0.029	-0.022	0.044	-0.029	-0.009	-0.106	0.059	0.049	0.049	0.023	0.216	-0.082	0.147	1	
(20) <i>IMR</i>	-0.075	-0.016	0.057	0.046	-0.012	0.036	-0.412	-0.133	-0.246	0.006	0.131	0.194	-0.132	-0.096	0.056	-0.036	0.129	-0.051	-0.064	1

All coefficients in bold are significant at at least the 5% significance level.

Refer to Appendix A for variable definitions.

All variables except *SUSTAIN*, *MINIMIZE*, *MIXED*, *CETR3*, *TAXEXPERT*, *FOR_OP*, *TLCF*, *SEGMENTS*, and *IMR* have been winsorized at the 1st and 99th percentiles.

TABLE 3
Equity Incentives - Seemingly Unrelated Bivariate Probit Regression

VARIABLES	(1) <i>SUSTAIN</i>	(2) <i>MINIMIZE</i>	(3) <i>SUSTAIN</i> Marginal Effects	(4) <i>MINIMIZE</i> Marginal Effects
<u>Managerial Incentives/Practices</u>				
<i>DELTA</i>	0.000048 3.281***	0.000004 0.270	0.000009 2.900***	-0.000002 -0.57
<i>VEGA</i>	-0.000135 -1.347	0.000353 3.857***	-0.000053 -2.41***	0.000097 3.88***
<u>Tax Planning</u>				
<i>SIZE</i>	0.125707 6.254***	-0.021287 -1.195	0.026248 6.14***	-0.012998 -2.69***
<i>LEVERAGE</i>	-0.543732 -2.755***	0.464511 2.722***	-0.141229 -3.37***	0.149398 3.22***
<i>PPE</i>	-0.351761 -1.847*	0.565438 3.380***	-0.111067 -2.72***	0.162933 3.58***
<i>FOR_OP</i>	-0.070417 -1.058	0.049359 0.830	-0.017329 -1.23	0.016792 1.01
<i>TLCF</i>	-0.150085 -3.200***	0.116323 2.790***	-0.038148 -3.88***	0.038440 3.28***
<i>SEGMENTS</i>	0.000754 0.070	-0.005026 -0.527	0.000522 0.22	-0.001304 -0.50
<u>Performance</u>				
<i>CV_PTBI</i>	-2.547150 -19.610***	0.242495 9.066***	-0.517806 -24.83***	0.216115 19.11***
<i>ROA</i>	5.675769 10.588***	-0.976894 -2.320**	1.186278 10.89***	-0.590808 -5.14***
<i>BTM</i>	-0.151029 -1.449	0.005820 0.070	-0.030066 -1.38	0.010673 0.47
<u>Investing/Reporting</u>				
<i>RD</i>	-7.190267 -9.559***	4.282788 7.757***	-1.729298 -11.31***	1.510307 10.02***
<i>DISC_ACC</i>	0.151220 0.507	-0.901739 -3.502***	0.096839 1.53	-0.234732 -3.36***
Constant	-0.467221 -0.670	-1.082376 -1.484		
LR Test of Independence	247.93	$p < 0.01$		
Observations	4,668	4,668		

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Each regression model is estimated with industry fixed effects (two-digit SIC code).

All variables except *SUSTAIN*, *MINIMIZE*, *FOR_OP*, *TLCF*, and *SEGMENTS* have been winsorized at the 1st and 99th percentiles.

The estimate of the correlation among the equation error terms is -0.523 and significant at the 1% level, consistent with the decision to focus on one dimension of tax strategy reducing the likelihood that the firm focuses on the other dimension of tax strategy.

Columns (3) and (4) present the bivariate marginal probability effects of each covariate on the probability of the dependent variable equaling one (i.e., of the firm adopting either a sustainable or minimization tax strategy).

TABLE 4
Selection Model – Probability of Purchasing Auditor Provided Tax Services

VARIABLES	TAXSERVICE
<i>TENURE</i>	0.063771 4.768***
<i>INSTOWN</i>	0.078532 0.458
<i>MERGER</i>	0.070882 0.880
<i>AUDIND</i>	-0.481663 -6.993***
<i>LNAUDITFEES</i>	0.312227 6.550***
<i>OPPORTUNITY</i>	-0.053622 -0.707
<i>SIZE</i>	0.071147 1.993**
<i>LEVERAGE</i>	0.282233 1.347
<i>PPE</i>	-0.299404 -1.234
<i>FOR_OP</i>	0.006800 0.083
<i>TLCF</i>	0.025187 0.479
<i>DISC_ACC</i>	1.134289 3.577***
<i>ATLCF</i>	0.046035 0.074
<i>EQINC</i>	-3.937517 -0.781
<i>RD</i>	0.838394 1.135
<i>BTM</i>	-0.201206 -1.995**
<i>ROA</i>	-0.540619 -1.016
<i>CASH</i>	0.176826 0.906
<i>DEP</i>	1.683956 1.139
<i>BIG4</i>	0.068127 0.478
<i>SECTIER</i>	-0.521715 -2.705***
Constant	-4.748557 -5.821***
Area under the ROC Curve	0.827
Observations	4,668

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The logistic regression model is estimated with industry (two-digit SIC code) and year fixed effects.

All variables except *TAXSERVICE*, *TENURE*, *MERGER*, *FOR_OP*, *TLCF*, *BIG4*, and *SECTIER* have been winsorized at the 1st and 99th percentiles.

TABLE 5
Expert Tax Advice - Seemingly Unrelated Bivariate Probit Regression

VARIABLES	(1) <i>SUSTAIN</i>	(2) <i>MINIMIZE</i>	(3) <i>SUSTAIN</i> Marginal Effects	(4) <i>MINIMIZE</i> Marginal Effects
<u>Managerial Incentives/Practices</u>				
<i>TAXEXPERT</i>	-0.043345 -0.859	0.103329 2.295**	-0.015067 -1.49	0.033290 2.86***
<u>Tax Planning</u>				
<i>SIZE</i>	0.131879 6.558***	-0.016928 -0.958	0.027607 6.69***	-0.011001 -2.37**
<i>LEVERAGE</i>	-0.590497 -3.006***	0.441209 2.595***	-0.149541 -3.57***	0.146366 3.16***
<i>PPE</i>	-0.359739 -1.872*	0.641811 3.809***	-0.121572 -2.97***	0.177861 3.90***
<i>FOR_OP</i>	-0.064533 -0.950	0.131134 2.157**	-0.023410 -1.69*	0.033251 1.99**
<i>TLCF</i>	-0.154694 -3.277***	0.091737 2.192**	-0.036813 -3.74***	0.033984 2.91***
<i>SEGMENTS</i>	-0.000226 -0.021	-0.004405 -0.462	0.000202 0.09	-0.001126 -0.43
<u>Performance</u>				
<i>CV_PTBI</i>	-2.524860 -19.487***	0.246306 9.167***	-0.515703 -24.72***	0.216624 19.11***
<i>ROA</i>	5.681235 10.598***	-0.936214 -2.223**	1.187869 10.88***	-0.591618 -5.15***
<i>BTM</i>	-0.168724 -1.629	-0.007282 -0.087	-0.032713 -1.51	0.005846 0.26
<u>Investing/Reporting</u>				
<i>RD</i>	-7.113835 -9.524***	4.517586 8.239***	-1.739652 -11.42***	1.562638 10.44***
<i>DISC_ACC</i>	0.132604 0.444	-0.942961 -3.660***	0.095920 1.51	-0.239901 -3.43***
<i>IMR</i>	-0.224824 -0.863	-0.325093 -1.434		
Constant	-0.370059 -0.515	-1.014474 -1.355		
LR Test of Independence	226.01	$p < 0.01$		
Observations	4,668	4,668		

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Each regression model is estimated with industry fixed effects (two-digit SIC code).

All variables except *SUSTAIN*, *MINIMIZE*, *TAXEXPERT*, *FOR_OP*, *TLCF*, *SEGMENTS*, and *IMR* have been winsorized at the 1st and 99th percentiles.

The estimate of the correlation among the equation error terms is -0.521 and significant at the 1% level, consistent with the decision to focus on one dimension of tax strategy reducing the likelihood that the firm focuses on the other dimension of tax strategy.

Columns (3) and (4) present the bivariate marginal probability effects of each covariate on the probability of the dependent variable equaling one (i.e., of the firm adopting either a sustainable or minimization tax strategy). These effects are estimated for the model without the inclusion of *IMR* because marginal effects cannot be estimated for factor variables.

TABLE 6
Information Transfer among Directors - Seemingly Unrelated Bivariate Probit Regression

VARIABLES	(1) <i>SUSTAIN</i>	(2) <i>MINIMIZE</i>	(3) <i>SUSTAIN</i> Marginal Effects	(4) <i>MINIMIZE</i> Marginal Effects
<u>Managerial Incentives/Practices</u>				
<i>INFO TRANSFER</i>	-0.071798 -2.535**	0.065220 2.590***	-0.018967 -3.12***	0.020735 3.02***
<u>Tax Planning</u>				
<i>SIZE</i>	0.160496 7.376***	-0.025554 -1.342	0.033414 7.25***	-0.016242 -3.14***
<i>LEVERAGE</i>	-0.564252 -2.869***	0.425070 2.502**	-0.142519 -3.41***	0.141053 3.05***
<i>PPE</i>	-0.349133 -1.836*	0.563370 3.369***	-0.110627 -2.71***	0.162492 3.57***
<i>FOR_OP</i>	-0.063737 -0.972	0.081308 1.386	-0.018450 -1.33	0.024582 1.49
<i>TLCF</i>	-0.156515 -3.344***	0.111278 2.672***	-0.039097 -3.98***	0.037637 3.21***
<i>SEGMENTS</i>	0.001677 0.155	-0.006729 -0.703	0.000832 0.36	-0.001788 -0.69
<u>Performance</u>				
<i>CV_PTBI</i>	-2.543110 -19.576***	0.246214 9.188***	-0.517596 -24.79***	0.217612 19.18***
<i>ROA</i>	5.597580 10.440***	-0.935224 -2.223**	1.168668 10.70***	-0.577440 -5.02***
<i>BTM</i>	-0.212660 -2.031**	-0.004659 -0.056	-0.041396 -1.89*	0.011874 0.52
<u>Investing/Reporting</u>				
<i>RD</i>	-7.108170 -9.531***	4.525143 8.255***	-1.733420 -11.41***	1.569134 10.48***
<i>DISC_ACC</i>	0.158554 0.531	-0.936518 -3.637***	0.100873 1.59	-0.243905 -3.48***
Constant	-0.749225 -1.067	-1.002369 -1.368		
LR Test of Independence	247.39	$p < 0.01$		
Observations	4,668	4,668		

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Each regression model is estimated with industry fixed effects (two-digit SIC code).

All variables except *SUSTAIN*, *MINIMIZE*, *FOR_OP*, *TLCF*, and *SEGMENTS* have been winsorized at the 1st and 99th percentiles.

The estimate of the correlation among the equation error terms is -0.520 and significant at the 1% level, consistent with the decision to focus on one dimension of tax strategy reducing the likelihood that the firm focuses on the other dimension of tax strategy.

Columns (3) and (4) present the bivariate marginal probability effects of each covariate on the probability of the dependent variable equaling one (i.e., of the firm adopting either a sustainable or minimization tax strategy).

TABLE 7
Managerial Incentives and Practices - Seemingly Unrelated Bivariate Probit Regression

VARIABLES	(1) <i>SUSTAIN</i>	(2) <i>MINIMIZE</i>
<u>Managerial Incentives/Practices</u>		
<i>DELTA</i>	0.164	0.022
	3.106***	0.478
<i>VEGA</i>	-0.035	0.167
	-0.667	3.564***
<i>TAXEXPERT</i>	-0.058	0.116
	-1.091	2.492**
<i>INFO TRANSFER</i>	-0.136	0.095
	-2.157**	1.732*
<u>Tax Planning</u>		
<i>SIZE</i>	0.443	-0.164
	6.173***	-2.642***
<i>LEVERAGE</i>	-0.155	0.138
	-2.626***	2.766***
<i>PPE</i>	-0.137	0.262
	-1.669*	3.726***
<i>FOR_OP</i>	-0.046	0.069
	-0.913	1.560
<i>TLCF</i>	-0.167	0.112
	-3.310***	2.548**
<i>SEGMENTS</i>	0.018	-0.035
	0.323	-0.718
<u>Performance</u>		
<i>CV_PTBI</i>	-4.864	0.449
	-19.550***	8.965***
<i>ROA</i>	0.732	-0.110
	10.424***	-2.039**
<i>BTM</i>	-0.117	0.039
	-1.719*	0.734
<u>Investing/Reporting</u>		
<i>RD</i>	-0.714	0.420
	-9.530***	7.788***
<i>DISC_ACC</i>	0.035	-0.189
	0.595	-3.748***
<i>IMR</i>	-0.058	-0.091
	-0.955	-1.753*
LR Test of Independence	224.43	$p < 0.01$
Observations	4,668	4,668

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Fully standardized coefficient estimates are reported.

Each regression model is estimated with industry fixed effects (two-digit SIC code).

All variables except *SUSTAIN*, *MINIMIZE*, *FOR_OP*, *TLCF*, *SEGMENTS*, and *IMR* have been winsorized at the 1st and 99th percentiles.

The estimate of the correlation among the equation error terms is -0.522 ($p < 0.01$), consistent with the decision to focus on one dimension of tax strategy reducing the likelihood that the firm focuses on the other dimension of tax strategy.

TABLE 8
Matched Firm Analysis - Logistic Regression

VARIABLES	(1) <i>SUSTAIN</i>	(2) <i>SUSTAIN</i>	(3) <i>SUSTAIN</i>	(4) <i>SUSTAIN</i>
<u>Managerial Incentives/Practices</u>				
<i>DELTA</i>	0.000145			0.000137
	3.557***			3.327***
<i>VEGA</i>	-0.000319			-0.000256
	-1.326			-1.016
<i>TAXEXPERT</i>		-0.263668		-0.288373
		-2.013**		-2.192**
<i>INFO TRANSFER</i>			-0.166700	-0.153725
			-2.348**	-2.062**
<u>Tax Planning</u>				
<i>SIZE</i>	-0.051347	0.040522	0.042139	0.062310
	-0.968	0.770	0.747	1.017
<i>LEVERAGE</i>	-1.842887	-2.199707	-2.021507	-1.956720
	-3.579***	-4.273***	-3.952***	-3.769***
<i>PPE</i>	-0.850532	-1.127290	-0.765173	-1.019456
	-1.716*	-2.272**	-1.551	-2.031**
<i>FOR_OP</i>	-0.217921	-0.378017	-0.183183	-0.357435
	-1.306	-2.234**	-1.113	-2.079**
<i>TLCF</i>	-0.220042	-0.175528	-0.244288	-0.176008
	-1.840*	-1.464	-2.056**	-1.453
<i>SEGMENTS</i>	-0.005683	-0.009122	-0.003918	0.000307
	-0.196	-0.314	-0.135	0.010
<u>Performance</u>				
<i>CV_PTBI</i>	-5.298183	-5.275233	-5.183406	-5.483688
	-14.941***	-15.038***	-14.923***	-15.120***
<i>ROA</i>	12.324620	12.088076	11.995359	11.992319
	8.037***	7.889***	7.886***	7.748***
<i>BTM</i>	-0.287562	-0.488319	-0.496883	-0.483626
	-1.022	-1.750*	-1.773*	-1.676*
<u>Investing/Reporting</u>				
<i>RD</i>	-17.759347	-17.865106	-17.456647	-17.975716
	-9.585***	-9.678***	-9.480***	-9.715***
<i>DISC_ACC</i>	1.019789	0.789634	0.908616	0.962777
	1.271	0.990	1.141	1.193
<i>IMR</i>		1.906891		1.888581
		2.674***		2.615***
Pseudo R-Square	0.328	0.328	0.324	0.335
Observations	1,998	1,998	1,998	1,998

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Each logistic regression model is estimated with industry fixed effects (two-digit SIC code).

The sample consists of 999 unique pairs of sustainable and minimization tax strategy firms that have been matched on *SIZE* within both industry (Fama French 17 classification) and year without replacement.

All variables except *SUSTAIN*, *MINIMIZE*, *FOR_OP*, *TLCF*, *SEGMENTS*, and *IMR* have been winsorized at the 1st and 99th percentiles.